

**DISCUSSION PAPERS
IN
ECONOMICS**

No. 2021/1

**DUAL-OBJECTIVE DONORS
AND INVESTMENT
ACROSS ECONOMIC REGIONS:
THEORY AND EVIDENCE**

KING YOONG LIM

CHUNPING LIU

JANUARY 2021

DISCUSSION PAPERS IN ECONOMICS

At Nottingham Business School, our Working Papers Series in Economics include a wide variety of approaches reflecting our research interests and academic expertise.

This paper is part of the series, ***Discussion Papers in Economics***.

All papers are work-in-progress, and all comments are therefore welcomed.

Earlier papers can be found at:

<https://www.ntu.ac.uk/research/research-at-ntu/academic-schools/research-at-nottingham-business-school/nbs-working-papers>

Please contact the Editor for enquiries concerning any of our Discussion Papers:

King Yoong Lim
Division of Economics
Nottingham Trent University
50 Shakespeare Street
Nottingham NG1 4FQ
UNITED KINGDOM

Email: king.lim@ntu.ac.uk

Tel: + 44 (0)115 848 6071

Dual-objective Donors and Investment across Economic Regions: Theory and Evidence

King Yoong Lim and Chunping Liu*

First completed draft: December 6, 2020

This version: February 7, 2021

Abstract

Based on a novel theoretical framework with a dual-objective donor giving grant (driven by benevolent objective) and loan aid (profitability motive), as well as a public private partnership-like investment structure and multi-region feedback mechanism, we estimate a panel structural VAR model of grant aid, loan aid, private consumption, and aggregate capital stock using data of nine economic communities/regions covering the 1961-2017 period. By allowing for cross-heterogeneity and interdependencies across regions, we identify empirical commonalities and differences across regions. Amidst a generally weak aid-investment link, we find loan aid to Granger cause capital accumulation, and among African regions it is the most (least) productive—in terms of within- and spillover effects—in the IGAD (ECOWAS) region. In addition, our results suggest deeper regional integration between the Sub-Saharan African regions than the collective Latin American & Caribbean regions and the two Southern Asian regions (Southeast Asia and South Asia) examined.

JEL Classification Numbers: F35, O40, P45

Keywords: foreign aid, donor objectives, investment, regional integration, structural VAR.

*Nottingham Business School, Nottingham Trent University. All views expressed in the article are our own. Useful feedback from Chuku Chuku, Allen Dennis, Peter Chacha Wankuru, and our departmental colleagues are greatly appreciated.

1 Introduction

Research on the subject of official development assistance (ODA) or foreign aid has spanned over half a century and been as eminent as the “White Man’s Burden” poem penned by Rudyard Kipling. In the post-War rebuilding era of the 1950s, aid disbursement consideration was interpreted as a requirement to meet saving gap in financing investment in recipient economies, courtesy of the linear output-capital relationship in the Harrod-Domar interpretation. This, and its extended two-gaps version (Chenery and Stout 1966), has been the dominant paradigm in driving an ‘aid-financed investment fetish’ in the 1960-70s period (Riddell 1987). Indeed, from aid being viewed as imperative to support development, to ODA being viewed as ineffective and arguably bring more trouble than help in developing economies (e.g. Easterley 2006, or the “Dead Aid” argument of Moyo 2009) towards the end of the century, the global aid landscape and its associated research agenda have evolved as the foreign and development policy paradigm shifts over the years. Indeed, the voluminous aid literature is best summarized as follows: a keyword search in Google Scholar for ‘aid effectiveness’ yields about 3.67 million results; ‘aid and investment’ yields about 3.3 million; ‘aid and regional integration’ yields 1.77 million.

In recent years, at the same time as the overarching objective of the OECD Development Assistance Committee (DAC) for the 2018-2022 period places greater emphasis on promoting development co-operation, globally ‘new’, non-disclosive emerging donors, including non-governmental private donors and the infamous China aid, have gained significance by capitalizing on the common perception that traditional aid is ineffective for growth and development in recipient economies (Lin and Wang 2017). The type of aid observed is also of more commercial nature and closely related to trade and investment (Kragelund 2008; Paulo and Reisen 2010). This phenomenon draws policymakers’ attention back to the intricate links between aid flows and investment, notably when there are conflicting donor objectives. Indeed, a keyword search for ‘aid and donor objectives’ would yield only about 160,000 results, underlying a relatively underexplored area of aid research. We contribute to the aid

and development literature by examining the dynamics of aid flows and investment under the presence of dual-objective donors who allocate ODA between traditional grant aid (driven by benevolent objective) and loan aid (driven by profitability motive). To motivate and clearly identify the interactions between the two types of aid and investment, we first develop a novel close-economy model with a dual-objective donor and a public-private partnership (PPP)-like investment mechanism, in which the model solutions can be summarized by a structural vector-autoregression (SVAR) system that intuitively links the dynamics of four key variables: traditional grant aid, loan aid, private consumption, and aggregate capital stock. In addition, to examine the growing policy issue of aid and South-South regional integration, the model economy is then naturally expanded to allow for interaction and feedback across multiple regions. To facilitate empirical estimation within a reasonable scale, we aggregate bilateral aid flows and capital stock data of different economies spanning the 1960-2017 period into nine economic communities/regions, so as to analyze the dynamic interactions of the different aid flows and investment both within and between the regions. Our study is closest in spirit to Juselius et al. (2014), who fit atheoretical cointegrated VAR model for 36 individual African economies and find evidence supporting an aid-driven growth in Africa. In comparison, our Bayesian-estimated panel SVAR at the regional level, by allowing for cross-heterogeneities and interdependencies, enables us to not just identify the drivers and dynamic relationship between variables in and across the different developing regions (using variance decomposition and impulse response analysis), but also distinguishes regions that have displayed aid-driven growth, or where aid is likely to have positive (negative) effects to capital investment. As such, although data limitation has restricted current aid research on the controversial Chinese aid to those based on media search (e.g. Dreher et al. 2018, 2019), our evidence (based on almost 60 years of DAC aid-based evidence) would allow for inference to be made in terms of which regions would likely benefit from the influx of commercial loan aid and which would not.

Our study is primarily related to three branches of literature on aid. First, at its theoret-

ical core, it concerns aid (in)effectiveness on capital investments—albeit at a macro-regional level—with the novel contribution being the interaction between the two types of aid and possibility of cross-regional spillovers.¹ On this branch of the largely empirical-dominated literature, the present consensus remains pretty much non-conclusive: as much as there is a large number of studies documenting aid ineffectiveness in driving economic growth in developing economies (e.g. Pedersen 2001; Easterley 2007; Roodman 2007; Rajan and Subramanian 2008), there are as many empirical studies that document aid effectiveness in promoting growth and development (e.g. Minoiu and Reddy 2010, Mekasha and Tarp 2013). Based on our interpretation, given the limitation of panel econometric techniques in fully exploiting the properties of time series data, traditionally the finding is that the aid-to-investment-to growth linkages are not very robust (Dollar and Easterley 1999), but at the same time the cumulative effects is believed to be stronger over the long run (Arndt et al. 2015). Given that the transmission mechanism from aid to growth is mostly likely through investment (Gomanee et al. 2005), our study contributes to this debate by finding that some regions do exhibit stronger positive aid-to-investment relationship, and some do not.

Second, our study is also related to the predominantly political economy-based literature on donor ideology and objectives. Drawing on a metaphor of *Merchant* and *Clergyman*, it has long been argued that donors’ aid decision tend to involve a persistent conflict between altruistic and self-interested motives, and any attempt to explain the heterogeneity in aid flows ought to have accounted for the possibility of conflicting objectives for the donors (Heinrich 2013; van Dam and van Dis 2014). Indeed, as argued by Morrissey (1993), donors’ tendency to mix aid and trade considerations would reflect the persistent trade-off between these two forces, and in most instances humanitarian motives appear to be insignificant

¹Some scholars, such as Riddell (2007), would argue forcefully that only country-based evidence ought to be used to examine aid effectiveness. While we agree that this is valid if country-specific lessons were the main focus, as it will become clearer, our regional empirical exercise allows for additional inferences to be made with regards to the aid-to-investment mechanism across regions; this obviously cannot be accomplished by standard cross-country regression.

compared to other considerations (Imbeau 1988). Empirically, studies such as Berthelemy (2006) and Kilby and Dreher (2010) provide some support to this argument. Our model formalizes this idea and provides some empirical evidence on the interactions between grant and loan aids.

Finally, our study also allows us to provide some evidence on the aid-for-trade and regional integration literature (Vijil 2014). Indeed, in instances where aid flows are explicitly linked to region-to-region trade negotiations, the effectiveness of aid would necessarily be reflected in wider regional integration (De Lombaerde and Mavrotas 2009). Nevertheless, little quantitative evidence on aid effectiveness in promoting regional integration (hence, regional absorption capacity) exists (Cadot et al. 2014). With the *African Continental Free Trade Area* (AfCFTA) being formally put in place in 2018, yet at the same time a large number of economies in the continent remains aid-dependent, we believe our dual-objective donor-driven growth framework would shed lights on some of these matters. To preview, for Sub-Saharan Africa, we find aid flows in the Kenya and Uganda-driven *Intergovernmental Authority on Development* (IGAD) region to exhibit the largest positive spillover effects to capital accumulation in other regions, followed by the *Economic Community of Central African States* (ECCAS). In contrast, the spillover effects associated with aid expansion into the *Economic Community of West African States* (ECOWAS) and to a lesser extent, *Southern African Development Community* (SADC), are negative.

The remainder of the article is structured as follows. Section 2 provides stylized facts and preliminary evidence that rationalizes our theoretical and empirical choice. Sections 3 and 4 motivate the theoretical model. Section 5 discusses our empirical strategy, followed by the presentation and analysis of estimation results in Section 6. Section 7 concludes the article.

2 Aid Flows & Capital Investment: Stylized Facts and Preliminary Evidence

Our preliminary analysis of the two major aid types of focus—traditional (in the form of grants) and loan aid (assumed to be driven by profitability/commercial objective)—is based on the ODA aid flows data compiled by the OECD, hence the focus on the 50-odd years period of 1960-2017. Globally, both the aggregate (as reported by both DAC and non-DAC donor economies) traditional and loan aid flows to all developing countries have steadily increased over the last half of the twentieth century up to the year 1991. Specifically, as seen in Figure 1, in per capita terms, grant aid flow increases from USD 0.93 million in 1970 to USD 7.5 million in 1991, while per capita loan aid flow increases from USD 0.77 million in 1970 to USD 3.6 million in 1991. Post 1991, per-capita flows of both aid types appear to have stagnated and on a declining trend to close the century; this period coincides with a global aid landscape that is increasingly fragmented (with the emergence of non-official private donors, some of which dramatically dubbed by as “rogue aid” by Naim 2009), and characterized by the well-documented increase in aid volatility (e.g. Pallage and Robe 2001; Bulir and Hamann 2003). Nevertheless, buoyed by renewed donor commitments made in the *Paris Declaration on Aid Effectiveness* in March 2005, since entering the 21st century the developing world has seen a resurgent in both aid flows again (Kharas 2007). Of note, both the DAC and the emerging non-DAC donors have increasingly opted for the use of project loan aid as an alternative means of development assistance, with the share of total loan aid increased from a 0.107 low in 2005 to 0.169 in 2017, and its share contributed by the non-DAC economies at 13.6 percent in 2017. Although this partly reflects some of the development in donors’ policy approach post-Paris Declaration (greater alignment to development strategies of recipient economies; more use of country programmatic approaches; quantitative targets’ setting), the growing significance of loan aid for project financing—be it direct or indirect through ‘blended’ finance—is unmistakable even when the infamous

Chinese aid is not accounted for.

Next, given that the large number of existing cross-country regression-based literature examining aid-growth relationship tend to find a shared similarity among economies with common colonial history, political alliances, institutions, or geographical proximity (e.g. Alesina and Dollar 2000; Rajan and Subramanian 2007; Jones 2015), we examine the interaction of the two types of aid across different economic regions/communities in the developing world. On aggregate, the *levels* of traditional aid and loan aid across the different economic regions appear to be positively correlated over time, as illustrated in Figure 2, which plots the *level* of total grants against the *level* of total loan aid received annually over the 1960-2017 period for the respective regions. Interestingly, the four Sub-Saharan African economic communities (ECCAS, ECOWAS, SADC, IGAD) appear to share similar patterns in terms of the co-movements of the two aid types over time. Further, a distinct, more ‘elongate’ pattern is observed for the two developing Asian regions (ASEAN, SAARC), whereas the Central America-based SICA and the South American grouping of economies have similar pattern over time. Despite these observations appear to be consistent with studies such as Burnside and Dollar (2000), who suggest that the level of aid allocation is mainly donor-driven, regional heterogeneity becomes more obvious when the *growth dynamics* are examined. Figure 3 plots the growth rates of both aid types over the same time period, and the positive co-movement between grant and loan aids largely disappear. Indeed, as some regions appear to display a vague positive pattern over time, other regions have shown a negative trade-off between the change in the two types of aid. Indeed, the two Asian regions (ASEAN, SAARC) appear to experience comparatively lower aid volatility over time, suggesting a different aid dynamics across these different economic communities.

Another common perception among developing policymakers and researchers alike is that, the rise of loan aid over time merely reflects a greater willingness of donor to participate in public-private partnership (PPP) investment projects in the recipient economies, as a concerted effort to respond to the well-documented ineffectiveness of traditional aid in promoting

growth in recipient economies. In terms of preliminary graphical evidence, we observe from the cross-country evidence in Figure 4 that, over a long run period, loan aid flows appears to be positively correlated with investment activities in the recipient economies. This positive association is noticed for both the level of aggregate capital stock and the more narrowly defined PPP capital stock, as compiled by the International Monetary Fund (IMF).

In addition to the stylized facts identified graphically, the basic time series properties of and between traditional aid (a_t^D), loan aid (a_t^L), recipient's physical capital stock (K_t^C), and private consumption (C_t) are examined across the nine economic communities/regions.² The upper panel of Table 1 reports the volatility of the log-level and first-difference measures of the four variables, and the lower panel reports the corresponding first-order autocorrelation of these variables. Overall, the statistics suggest the two types of aid to exhibit typical properties of macroeconomic time series: some degree of persistence due to the presence of a trend, integrated of order one, and difference stationary. Specifically, trend volatility of the two types of aid is not significantly different from consumption and capital stock, with the above-average volatility associated with the Sub-Saharan African communities of ECOWAS, SADC, and IGAD consistent with those documented in the empirical aid literature (for instance, Kumi et al., 2017). Next, loan aid, followed by traditional aid, does exhibit relatively more dynamics than the two 'domestic' variables. Nevertheless, the persistence of loan aid is comparatively lower, with its average first-order autocorrelation of 0.621 being much lower than the other three variables. These, couple with the predominantly negative first-order autocorrelation values observed for both aid types across the different regions, suggest that the aid allocation process (between the two types of aid) is likely to exhibit dynamic reallocation influenced by a mixture of different factors and possibly, conflicting objectives.

Further, Table 2 reports the contemporaneous correlations between the nine developing economic regions/communities. Positive correlation values between the levels of traditional

²Due to a large amount of missing data for the non-DAC donors in the early years, as well as the fact that many of the non-DAC donors, including China, do not report their aid activities, for the time series analysis we use only the aid data for the DAC donors.

aid across all nine regions are consistently observed for traditional aid, albeit at different magnitudes. Of note, the four Sub-Saharan African communities have high correlation with each others, with the 0.886 between the Central African bloc of ECCAS and the Horn of Africa/East Africa-based IGAD being the smallest. Similarly, the two non-MERCOSUR country blocs in Latin America & Caribbean, as well as the two developing Asian regions, register high correlation values. This high correlation of traditional aid is nonetheless not observed for their growth rates, as some regions with close geographical proximity even register negative correlation. Indeed, the correlation across regions for both the *levels and growth rates* of loan aid are also much smaller, and the corresponding growth rates differ, even for the four African economic regions (unlike for the traditional aid). Collectively, these suggest that there are different dynamics associated with the two types of aid, as well as their interactions with the domestic investment and consumption activities, across the different economic regions. Indeed, between the different regions, especially those within the same continents where large-scale, multi-regional investment project facilities are in place, there may be cross-region spillover and feedback between aid and investment flows.

Finally, to provide empirical basis in guiding our theoretical specification, we also formally test for pairwise Granger causality on the first difference of the four variables (a_t^D , a_t^L , K_t^C , C_t) using the panel data of the nine regions. Specifically, Table 3 reports the Z-bar statistics and the corresponding p-value for the series of pairwise Dumitrescu and Hurlin (2012) Granger non-causality tests. In sum, for at least one of the panels, the test results suggest that the growth of loan aid Granger-cause both the growth of aggregate capital stock and consumption; the growth of consumption and traditional aid mutually Granger-cause; and consumption growth Granger-cause growth in capital stock. These preliminary results are not only consistent with the common perception of the relationship between traditional aid and consumption, but also reinforces the earlier graphical observation that aid loan flows do potentially lead to higher investment activities in the emerging market regions.

3 Theoretical Framework

To provide a theoretical basis to our empirical specification, we first present the theoretical model of a closed economy where, in the presence of a dual-objective optimizing donor and public-private partnerships (PPP) investment arrangement involving the three parties of private sector, domestic government, and foreign sectors, the dynamic system characterizing the model solutions can be summarized by a structural VAR system of four variables: traditional aid (a_t^D), loan aid (a_t^L), recipient's physical capital stock (K_t^C), and private consumption (C_t). Specifically, the foreign donor provides two forms of development assistance to the economy. The total development assistance made by the donor to the recipient economy in any period t is $a_t = a_t^D + a_t^L$, where a_t^D is traditional aid in the form of grant and a_t^L is loan aid. In each period, the former is modelled as being transferred by the donor to the government, who would then transfer it to the household, whereas the latter is modelled stylistically as investment flows/inputs contributed to physical capital investments under a PPP investment arrangement. The traditional aid form is meant for pure benevolent objectives, whereas the loan aid is interpretable as driven by profitable motive. Output prices in the economy are normalized to unity, which means all variables and rates are modelled in real terms.

Having illustrated the dynamic causation and linkages between these four key variables, the single economy model can then be easily modified and expanded to account for mutual feedback and spillover across multiple regions $j = 1, \dots, J$. In other words, it presents the framework of a $4 \times J$ SVAR system for the subsequent empirical investigation.

3.1 Physical capital accumulation

To model an investment arrangement under the PPP structure commonly observed in developing economies (where the local government often co-finance PPP investment projects with external financiers, such as foreign governments and international development corporation),

the equation of motion for physical capital accumulation in the economy is specified as

$$K_{t+1}^C = (I_t)^\varpi (a_t^L)^{\kappa(1-\varpi)} (G_t^G)^{(1-\kappa)(1-\varpi)} + (1-\delta)K_t^C, \quad (1)$$

where K_t^C is the PPP-capital stock, I_t the investment made by the private household, a_t^L the commercial aid made by the foreign donor, G_t^G the contribution to PPP-investment made by the government, $\varpi, \kappa \in [0, 1]$ are the investment elasticities that denote the relative shares, δ is the depreciation rate. The “public” component of the PPP is therefore co-financed by the local government and the foreign donor. In each period t , the three contributors make their contributions, which are compensated by their respective gross returns: $\varpi(1+r_{t+1}^C)K_{t+1}^C$ for the private household, $\kappa(1-\varpi)(1+r_{t+1}^C)K_{t+1}^C$ for the foreign donor, $(1-\kappa)(1-\varpi)(1+r_{t+1}^C)K_{t+1}^C$ for the government.³ These capital returns are transferred to the respective budgets at the end of the period and hence appear in the respective budget constraints only in the next period. For empirical implementation, although conceptually structured as a PPP framework, this specification is general enough that it can also be easily interpretable at the macro level in which K_t^C is a single aggregate capital stock that includes all private investments, public investments, and ‘actual’ PPP investments.

3.2 Government

The government taxes effective wage income, $w_t Z_t N_t$, where Z_t is labor-augmenting productivity, N_t is labor units supplied by the household, at a rate $\tau \in (0, 1)$, and receives the gross returns from co-investing in PPP-capital. In addition, in each period t , the government receives traditional aid, a_t^D , in the form of grants, from the foreign donor, and assumed to pay out a fraction, $\chi \in (0, 1)$, as cash transfer to the household, χa_t^D . This therefore allows for some degree of imperfection to be introduced in the aid disbursement process. The gov-

³The gross returns can easily be derived by visualizing the set-up of a PPP-capital assembly firm (special purpose vehicle) making zero profits. It maximizes a period-specific objective function, $\Pi_t = (1+r_{t+1}^C)K_{t+1}^C - mc_t(I_t + a_t^L + G_t^G)$, where mc_t is the marginal cost of a unit of fund contributed. Solving this unconstrained maximization problem for I_t , a_t^L , G_t^G would yield the respective gross compensations stated.

ernment consumes G_t^O amount of final goods, and incurs G_t^G as a co-investor in PPP-capital. We assume $G_t^G = v_L a_t^L$, $v_L \in (0, 1)$, $\forall t$, which implies that for PPP-investments the government is only contributing a fraction of each dollar that the foreign donor is contributing. Given that public debt dynamics are a peripheral issue to the main objective of this study, to minimize unnecessary model complexity, we assume a government that maintains a balanced budget and does not borrow. The budget constraint of the government is given by

$$\tau w_t Z_t N_t + (1 - \kappa)(1 - \varpi)(1 + r_t^C)K_t^C + a_t^D = \chi a_t^D + G_t^G + G_t^O. \quad (2)$$

3.3 Household

To simplify population dynamics, we assume a representative household that is endowed with a labor unit, N_t , which is fully spent in working. The representative household chooses a sequence of consumption and investment, $\{C_{t+s}, I_{t+s}\}$, for $s = 0, 1, \dots, \infty$, so as to maximize lifetime utility:

$$U_t = \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \left[\frac{(C_{t+s})^{1-\varsigma^{-1}}}{1 - \varsigma^{-1}} \right], \quad (3)$$

subject to the budget constraint,

$$(1 - \tau)w_t Z_t N_t + \varpi(1 + r_t^C)K_t^C + \chi a_t^D = C_t + I_t, \quad (4)$$

where $\beta > 0$ is the discount rate, $\varsigma > 0$ the intertemporal elasticity of substitution, $w_t Z_t N_t$ is effective labor income, $\tau \in (0, 1)$ is the constant labor tax rate, $\varpi(1 + r_t^C)K_t^C$ is the gross returns received from contributing to PPP in the previous period, and χa_t^D is the cash transfer from the government, which is a fraction $\chi \in (0, 1)$ of the traditional aid/grant provided by the donor, a_t^D . As derived in Appendix A, solving the intertemporal problem

yields the household optimality condition,

$$(1 + r_{t+1}^C) = \left(\frac{C_{t+1}}{C_t}\right)^{\varsigma^{-1}} \left[\frac{K_{t+1}^C}{K_t^C} - (1 - \delta) \right]^{\frac{1-\varpi}{\varpi}} \frac{(\Theta_1)^{-1}}{\beta \varpi^2} \left(\frac{a_t^L}{K_t^C}\right)^{\frac{\varpi-1}{\varpi}} - \beta(1 - \delta) \left[\frac{K_{t+2}^C}{K_{t+1}^C} - (1 - \delta) \right]^{\frac{1-\varpi}{\varpi}} \frac{(\Theta_1)^{-1}}{\beta \varpi^2} \left(\frac{a_{t+1}^L}{K_{t+1}^C}\right)^{\frac{\varpi-1}{\varpi}}, \quad (5)$$

where $\Theta_1 = v_L \hat{\{[(1 - \kappa)(1 - \varpi)]/\varpi\}}$. (5) shows that, in an economy where investments are made under PPP agreement, the required rate of return that corresponds to optimal intertemporal allocation of consumption would depend not only on the returns and rate of capital accumulation over time, but also the contributions made by the foreign donor and the government.

3.4 Foreign Donor and Development Finance

To finance both types of development assistance, it is assumed that the foreign donor commits an exogenous amount, $(1 - \Theta)L_t$ per capita (in effective term, hence adjusted by Z_t), to the recipient economy in each period t . We also assume all the returns received from PPP-investment in the previous period are recommitted back to the pool of funds made for the recipient economy. The donor's budget constraint is therefore

$$\underbrace{(1 - \theta)L_t Z_t N_t}_{\text{Total allocation made in period } t} + \underbrace{\kappa(1 - \varpi)(1 + r_t^C)K_t^C}_{\text{Gross returns of PPP-investment, re-added to the total allocation}} = \underbrace{a_t^L + a_t^D}_{\text{Allocated to traditional grants \& commercial aid}}. \quad (6)$$

⁴First, underlying the framework but not explicitly modelled, $(1 - \Theta)L_t$ can be interpreted as an agreement reached in the beginning of the period, $t = 0$, where the donor sets aside an initial sum, L_t , to be distributed in equal sum each period. Specifically, given a constant sum, C , discounted using the donor's discount rate, Θ , we have $C(1 + \Theta + \Theta^2 + \dots + \Theta^T) = L_t$. Given infinitely-lived agents, $T \rightarrow \infty$, we therefore have $C = L_t(1 - \Theta)$. Second, the "re-commitment" specification is made to ensure all resource flows in the model are properly accounted for. Indeed, the alternative of a specification without this component would have made the donor's problem a lot more easier to solve for.

Given the dual objective, the donor's intertemporal optimization problem involves selecting sequences of traditional aid, a_{t+s}^D , and loan aid, a_{t+s}^L , subject to (6), so as to maximize the total discounted utility streams, taking the government's policy variables (including G_t^G), private investments, initial conditions, existing stocks, prices, and rates as given:

$$\{a_{t+s}^D, a_{t+s}^L\}_{s=0}^{\infty} = \arg \max \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \left[\frac{\{(a_{t+s}^D)^{\gamma(1-\varsigma^{-1})} [\kappa(1-\varpi)(1+r_{t+s+1}^C)K_{t+s+1}^C]^{(1-\gamma)(1-\varsigma^{-1})}\} - 1}{1 - \varsigma^{-1}} \right], \quad (7)$$

where $\gamma \in (0, 1)$ measures the relative preference of the donor with respect to her benevolent and profit-maximizing objectives. Given that the donor's objective with traditional aid is ultimately to increase the income transfer made to household, a similar CES-functional form to the specification of (3) is adopted. Hence, the choice variable associated with donor's objective is a_t^D , and not C_t , as in (3). Similar consideration also leads us to specify the same discount rate, β , and the same intertemporal elasticity of substitution, ς , to those of the household. As shown in Appendix A, solving the donor's intertemporal optimization problem yields the donor's optimality condition,

$$\begin{aligned} & \frac{(1-\gamma)}{\gamma} \frac{a_t^D}{a_t^L} \kappa(1-\varpi) \left[1 - (1-\delta) \frac{K_{t+1}^C}{K_t^C} \right] \\ &= 1 - \frac{\beta(1+r_{t+1}^C)}{[\kappa(1-\varpi)]^{-2}} \frac{[\frac{K_{t+1}^C}{K_t^C} - (1-\delta)]}{a_t^L/K_t^C} \left(\frac{a_{t+1}^D}{a_t^D} \right)^{\gamma(1-\varsigma^{-1})-1} \left[\frac{(1+r_{t+2}^C)K_{t+2}^C}{(1+r_{t+1}^C)K_{t+1}^C} \right]^{(1-\gamma)(1-\varsigma^{-1})}, \end{aligned} \quad (8)$$

which contains information on both intra- and inter-temporal allocation of development finance. (8) implies that the inter-temporal allocation of development finance between traditional and loan aid, having accounted for intra-temporal allocation between the two types of aid, would depend on the relative growth rates of both types of aid, as well as the standard rates of return consideration as contained in a standard Euler equation.

3.5 Production and Aggregate Resource Constraint

There is a perfectly competitive representative firm producing final good, Y_t , using labor input, N_t , and the physical capital stock produced under the PPP arrangement, K_t^C , taking initial stocks, price (normalized to one), rates of return $(w_t, 1 + r_{t+1}^C)$, as well as the Cobb-Douglas production technology,

$$Y_t = A_t(K_t^C)^\alpha(Z_t N_t)^{1-\alpha}, \quad (9)$$

where $\alpha \in (0, 1)$ is the elasticity with respect to PPP-capital stock, A_t is the Hicks-neutral technological level (with an initial level, A_0), as given. The variation of technology is assumed to be exogenous, stochastic, and has zero mean, with the evolution of its log-deviations ($\hat{A}_t = A_t - \tilde{A}$, $\forall t$, where $\tilde{A} = A_0$ is its steady-state level) assumed to follow $\hat{A}_{t+1} = \rho_A \hat{A}_t + \varepsilon_t$ $\forall t$, where $\rho_A \in (0, 1]$, and $\varepsilon_t \sim N(0, \sigma_A^2)$. Further, we also follow emerging market-based studies such as Aguiar and Gopinath (2007), García-Cicco et al. (2010), and specify a labor-augmenting productivity component, Z_t , to provide a ‘catch-all’ source of the cumulative products of past ‘growth shocks’,

$$\begin{aligned} Z_t &= \exp(g_t)Z_t = \Pi_{s=0}^t \exp(g_s), \\ g_t &= (1 - \rho_g)\tilde{g} + \rho_g g_{t-1} + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_g^2), \end{aligned} \quad (10)$$

where $|\rho_g| < 1$, and ϵ_t is independent and identically distributed with mean zero and constant variance, σ_g^2 . The term $\tilde{g}$ is the long-run mean growth rate of productivity, and in a balanced growth equilibrium would equal the steady-state growth rate.⁵ Although trend volatility is not a huge concern with our data, the different interactions of this additional cumulative source of randomness with the parameters in the different sectors over time [note that Z_t appear in (2), (4), and (6) too] provides a convenient ‘shorthand’ way to interpret the

⁵Likewise, if past “shocks” do not influence present labor-augmenting productivity ($\rho_g = 0$), then Z_t would just grow at a constant rate of $\tilde{g}$.

realization of different empirical shocks to be examined in our empirical analysis later.⁶

In each period t , the final-good firm maximizes profits, $\pi_t = Y_t - w_t N_t - (1 + r_{t+1}^C) K_t^C$, which yields first-order conditions:

$$(1 + r_{t+1}^C) = \alpha \frac{Y_t}{K_t^C}, \quad w_t = (1 - \alpha) \frac{Y_t}{Z_t N_t}. \quad (11)$$

To close the model, first, we know that the domestic absorption in each period consists of $C_t + I_t + G_t^O + a_t^L + G_t^G$. Next, the production of final good in each period, Y_t , is adjusted by the flows associated with the donor's activities to obtain the net amount that is available to meet the domestic absorption, $Y_t - \kappa(1 - \varpi)(1 + r_t^C) K_t^C + (a_t^D + a_t^L)$. Equating the two allows us to derive the aggregate resource constraint of the economy,⁷

$$Y_t - \kappa(1 - \varpi)(1 + r_t^C) K_t^C = C_t + I_t + G_t^O + G_t^G - a_t^D. \quad (12)$$

4 Dynamic System and Extension to Multiple Regions

4.1 Model Dynamics

With the labor units normalized to unity, $N_t = \bar{N} = 1$, the dynamic system characterizing the model solutions is obtained by applying log-linearization technique around a steady-state equilibrium of the model. Denoting the steady-state equilibrium values of the endogenous variables by *tildes*, and the log-deviations (from the steady-state levels) of variables with a *hat*, as derived in Appendix B, the dynamics of the theoretical model can be summarized by a SVAR system of the growth rates of four endogenous variables:

⁶Alternatively, we can introduce four different sources of innovations in place of the Z_t in (2), (4), (6), and (9) instead. In a largely empirical estimation-driven study, we are in the views that there is very little to gain in terms of insights, but at the cost of additional complexity for the readers.

⁷A quick verification of this identity can be made by combining the household resource constraint, (4), the government's budget constraint, (6), and (12). These would yield $w_t N_t + (1 + r_t^C) K_t^C = Y_t$, which would be the national income identity in a standard setting without PPP and foreign donor.

$$\begin{pmatrix} \hat{C}_{t+1} - \hat{C}_t \\ \hat{K}_{t+2}^C - \hat{K}_{t+1}^C \\ \hat{a}_{t+1}^D - \hat{a}_t^D \\ \hat{a}_{t+1}^L - \hat{a}_t^L \end{pmatrix} = \begin{pmatrix} 0 & \Omega_{H,2} & \Omega_{H,4} & \Omega_{H,6} \\ \Omega_{F,6} & 0 & \Omega_{F,2} & \Omega_{F,4} \\ \Omega_{G,6} & \Omega_{G,2} & 0 & \Omega_{G,4} \\ \Omega_{E,6} & \Omega_{E,2} & \Omega_{E,4} & 0 \end{pmatrix} \begin{pmatrix} \hat{C}_{t+1} - \hat{C}_t \\ \hat{K}_{t+2}^C - \hat{K}_{t+1}^C \\ \hat{a}_{t+1}^D - \hat{a}_t^D \\ \hat{a}_{t+1}^L - \hat{a}_t^L \end{pmatrix} \quad (13)$$

$$+ \begin{pmatrix} \Omega_{H,1} & \Omega_{H,3} & \Omega_{H,5} & \Omega_{H,7} \\ \Omega_{F,7} & \Omega_{F,1} & \Omega_{F,3} & \Omega_{F,5} \\ \Omega_{G,7} & \Omega_{G,3} & \Omega_{G,1} & \Omega_{G,5} \\ \Omega_{E,7} & \Omega_{E,3} & \Omega_{E,5} & \Omega_{E,1} \end{pmatrix} \begin{pmatrix} \hat{C}_t - \hat{C}_{t-1} \\ \hat{K}_{t+1}^C - \hat{K}_t^C \\ \hat{a}_t^D - \hat{a}_{t-1}^D \\ \hat{a}_t^L - \hat{a}_{t-1}^L \end{pmatrix} + \begin{pmatrix} \Gamma_{H,t} + \varepsilon_t \\ \Gamma_{F,t} + \varepsilon_t \\ \Gamma_{G,t} + \varepsilon_t \\ \Gamma_{E,t} + \varepsilon_t \end{pmatrix},$$

where $\Gamma_{i,t}$, $i = H, F, G, E$, denote the variable-specific random shocks, and ε_t the common random errors component.

4.2 Introducing mutual feedback and spillover across multiple regions

With the 4×1 SVAR form derived for the benchmark model, we are now in a position to expand the framework to a South-South model that includes multiple regions/economies. Specifically, to extend the benchmark framework to an open-economy context, the following modifications are made. First, final good production function will need to be rewritten to:

$$Y_t = A_t [(K_t^C)^\alpha (Z_t N_t)^{1-\alpha}]^\vartheta (\Upsilon_t^K)^{1-\vartheta}, \quad (14)$$

where $\Upsilon_t^K = \sum_j \Upsilon_{j,t}^K$ is the sum of imported capital goods from J regions, which would then yield a dynamics of:

$$\begin{aligned} \hat{Y}_t - \hat{Y}_{t-1} = & \alpha \vartheta (\hat{K}_t^C - \hat{K}_{t-1}^C) + (1 - \alpha) \vartheta (\hat{Z}_t - \hat{Z}_{t-1}) \\ & + (1 - \vartheta) (\hat{\Upsilon}_t^K - \hat{\Upsilon}_{t-1}^K) + (\hat{A}_t - \hat{A}_{t-1}), \end{aligned} \quad (15)$$

where $\hat{\Upsilon}_t^K - \hat{\Upsilon}_{t-1}^K = (\hat{\Upsilon}_{1,t}^K - \hat{\Upsilon}_{1,t-1}^K) + (\hat{\Upsilon}_{2,t}^K - \hat{\Upsilon}_{2,t-1}^K) + \dots + (\hat{\Upsilon}_{J,t}^K - \hat{\Upsilon}_{J,t-1}^K)$.

Likewise, the intertemporal-optimized household consumption, C_t , in (3) would need to be rewritten as a composite final consumption good that consists of the domestically produced final good and all imported foreign final goods (for consumption) from other economic regions ($\Upsilon_t^C = \sum_j \Upsilon_{j,t}^C$, $j = 1, \dots, J$), which then after log-linearization of the Euler equation, would give a dynamic form [similar to (B1) in Appendix B] linking the growth rates of the different external economies, $\hat{\Upsilon}_{j,t+1}^C - \hat{\Upsilon}_{j,t}^C$, to each others, as well as the growth rates of the four key domestic variables ($C_t, I_{t+1}^C, a_t^D, a_t^L$) established in the benchmark model.

The domestic aggregate resource constraint, (12), would then naturally be rewritten into an open economy version:

$$Y_t = \mu(1 - \varpi)(1 + r_t^C)K_t^C = C_t + I_t + G_t^O + v_L a_t^L - a_t^D + (X_t - \Upsilon_t), \quad (16)$$

where total imports and exports are given by $\Upsilon_t = \Upsilon_t^K + \Upsilon_t^C$, and $X_t = \sum_j X_{j,t}^K + \sum_j X_{j,t}^C$, respectively, with $X_{j,t}^K$ and $X_{j,t}^C$ refer to the quantity (in terms of domestically produced final good) exported for production and consumption purposes in region j . Assuming symmetry, in that all the other economic regions, $j = 1, \dots, J$, have the same domestic economic structure as in the benchmark model, let Δ denotes the difference of variables, the SVAR system characterizing the benchmark model solutions, (13), can then be easily expanded to a larger, $4 \times J$ SVAR system, $\Phi = [\Delta \hat{C}_{j,t} \quad \Delta \hat{K}_{j,t+1}^C \quad \Delta \hat{a}_{j,t}^D \quad \Delta \hat{a}_{j,t}^L \quad \dots \quad \Delta \hat{C}_{J,t} \quad \Delta \hat{K}_{J,t+1}^C \quad \Delta \hat{a}_{J,t}^D \quad \Delta \hat{a}_{J,t}^L]'$ for a total of J economic regions.

5 Empirical Model and Data Treatments

5.1 Panel SVAR model

For the empirical estimation of (13), we apply panel VAR estimation to our SVAR model by combining economies into different regions to examine for commonalities and differences in regional dynamics over time (see Canova and Ciccarelli, 2013), allowing for cross-heterogeneities and interdependencies. For each unit i in a standard VAR, there are n endogenous variables, p lags over T periods and m exogenous variables, which is assumed to be common across units. If we stack all the N units, it becomes the following:

$$Y_t = \sum_{k=1}^p A_t^k Y_{t-k} + C_t X_t + \varepsilon_t, \quad (17)$$

where Y_t is a $Nn \times 1$ vector of endogenous variables, A_t^k is a $Nn \times Nn$ matrix of coefficients, C_t is $Nn \times m$ matrix and ε_t is a $Nn \times 1$ vector of random errors. We reformulate (17) in compact form, it becomes:

$$Y_t' = x_t B_t + \varepsilon_t', \quad (18)$$

where $x_t = \underbrace{(Y_{t-1}' \dots Y_{t-p}' X_t')}_{Nn \times h}$ and $B_t = \begin{pmatrix} (A_t^1)' \\ \vdots \\ \vdots \\ A_t^{(p)'} \\ (C_t)' \end{pmatrix}$. If we vectorize Equation (18), we get:

$$Y_t = \bar{x}_t \beta_t + \varepsilon_t, \quad (19)$$

with $\bar{x}_t = I_{Nn} \otimes x_t$ and $\beta_t = \text{vec}(B_t)$. Therefore, the number of coefficients to be estimated is $h = Nn(Nnp + m)$, which is problematic in reality, as there are only Nn observations

available. Considering this situation, Canova and Ciccarelli (2013) use a structural factor approach, where the coefficient vector β_t , which is a reduced-form representation of the transmission mechanism, is then assumed to be a linear combination of a set of structural factors. If we only consider the static case instead of time-varying approach, the factor model can be specified as:

$$\beta = \Xi_1\theta_1 + \Xi_2\theta_2 + \Xi_3\theta_3 + \dots + \Xi_r\theta_r, \quad (20)$$

where $\theta_{k,k=1,\dots,r}$ are vectors containing loadings of the ‘common components’, ‘unit-specific components’, ‘variable-specific components’ and exogenous variables, respectively, for each cross-sectional units; $\Xi_{k,k=1,\dots,r}$ are matrices with entries equalling either 0 or 1, which map the loadings with elements in y_t according to the structural factor restrictions. By doing so, this significantly reduces the number of coefficients to be estimated. For example, if we have 4 regions and 4 variables in the panel, with 1 lag order of 1 exogenous variable, we only need to estimate 9 parameters instead of 272 (16×17). Alternatively, we can write (20) as $\beta = \Xi\theta$, which if substituted into (19), yields:

$$Y_t = \tilde{x}_t\theta + \varepsilon_t, \text{ where } \tilde{x}_t = \bar{x}_t\Xi. \quad (21)$$

5.2 Priors and posteriors

Our panel SVAR model is estimated by Bayesian estimation. The parameters of interest are $\theta = \{\theta_1, \theta_2, \theta_3, \dots, \theta_r\}$, $\varepsilon_t \sim N(0, \sigma\tilde{\Sigma}_{uu})$, where $\tilde{\Sigma}_{uu}$ is the variance-covariance matrix and σ is a constant. The posteriors of θ , σ and $\tilde{\Sigma}_{uu}$, is calculated based on a classical Bayes rule, which combines the prior distributions of θ , σ and $\tilde{\Sigma}_{uu}$ and the likelihood function $f(Y|\theta, \sigma, \tilde{\Sigma}_{uu})$. Based on (21), the likelihood function is specified as:

$$f(Y|\theta, \sigma, \tilde{\Sigma}_{uu}) \propto (\sigma)^{-TNn/2} |\tilde{\Sigma}_{uu}|^{-T/2} \prod_{t=1}^T \left\{ -\frac{1}{2} \sigma^{-1} (Y_t - \tilde{x}_t \theta)' \tilde{\Sigma}_{uu}^{-1} (Y_t - \tilde{x}_t) \right\}. \quad (22)$$

The prior distributions of θ , σ and $\tilde{\Sigma}_{uu}$ are specified as the following equations:

$$\pi(\theta|\theta_0, \Theta_0) \propto \exp \left(-\frac{1}{2} (\theta - \theta_0)' \Theta_0^{-1} (\theta - \theta_0) \right) \quad (23)$$

$$\pi(\tilde{\Sigma}_{uu}) \propto |\tilde{\Sigma}_{uu}|^{-(Nn+1)/2} \quad (24)$$

$$\pi(\sigma) \propto \sigma^{-\frac{\alpha_0}{2}} \exp \left(-\frac{\delta_0}{2\sigma} \right), \quad (25)$$

where θ_0 is normally set to be zero and Θ_0 is specified as a diagonal matrix with large values to produce an uninformative prior, $\tilde{\Sigma}_{uu}$ is assumed to be an uninformative prior, and σ follows an inverse Gamma distribution with shape α_0 and scale δ_0 . With all this information, we can calculate the posterior distribution using Gibbs sampling. The models are estimated on chains with 100,000 draws: 2000 blocks of 50 draws were made and the last draw for each block is retained after discarding the first 50,000. This means that a total of 1000 draws is used to conduct posterior inference. α_0 and δ_0 are suggested to be 0.001 for the static factor model (Dieppe et al. 2018).

5.3 Identification of shocks

Although there is increasing use of DSGE models, structural VAR models continue to play an important role in applied macroeconomic studies. The identification of structural shocks in SVAR requires additional assumptions based on economic theories and prior knowledge, making economic interpretations more meaningful. There are a few identification schemes in the literature, such as Cholesky decomposition (zero short-run restrictions, i.e. Sims 1980),

sign restrictions (Uhlig 2005), zero long-run restrictions (Blanchard and Quah 1989), etc. However, our paper concentrates on Cholesky decomposition, as there is limited information or theory to support sign restrictions and long-run restrictions between loan and traditional grant aids in reality. As discussed in the literature, the ordering of variables is important if we use Cholesky decomposition. For the benchmark empirical set-up, consistent with (13) and that there isn't quarterly ODA data for us to have a longer T to work with, we estimate the SVAR model with one lag. With Cholesky identification, both the ordering of the four variables and the economic regions matter. Our structural ordering of the former is based on the panel Granger causality test results in Table 3, which suggest an ordering of loan aid (a_t^L), traditional grant aid (a_t^D), consumption (C_t), and then aggregate capital stock (K_t^C). For the regions, we estimate the SVAR model based on three regional integration themes. First, for the benchmark estimation the focus is on Sub-Saharan Africa, which is timely given the consistent effort of regional policymakers in pushing for further regional integration. Specifically, for the benchmark case with $J = 6$, we adopt a region ordering of developing Asia (which combines ASEAN and SAARC), Latin America & Caribbeans (LAC; which combines CSME-SICA, CAN, and MERCOSUR), followed by SADC, IGAD, ECCAS, and ECOWAS. This specification choice allows us to dwell deeper into the policy issues of inter-regional integration among the four African communities, as well as the relatively exogenous—albeit in a fully endogenous SVAR system—impacts from the broader regions of developing Asia and LAC on Sub-Saharan Africa. Moreover, we also explore two additional sets of region ordering: (i) developing Asia, Africa (which combines SADC, IGAD, ECCAS, ECOWAS), MERCOSUR, CAN, CSME-SICA; (ii) LAC, Africa, SAARC, ASEAN. The former would cast brighter lights onto regional integration issues within Latin America, whereas the latter focuses on the two developing Asian regions. For these two latter sets with a smaller J (hence additional degree of freedom available), due to the inclusion of the entire Africa as an individual region, and partly for robustness consideration, we would estimate the SVAR model with two lags instead so as to allow a lengthier period of time for any transmission

mechanism to take place.

5.4 Data

The sample period covered is **1960-2017**. As mentioned, for the region-specific variables of traditional aid (a_t^D) and loan aid (a_t^L), we construct the data by aggregating the annual disbursement flows of the total grants and total ODA gross loans received by all the economies grouped within the respective regions. Aggregation by economic regions allows us to minimize any potential idiosyncrasy associated with country-specific noise and misreporting, but still keep sufficiently unique characteristics of policy significance, given that donors' aid choices tend to be explainable by colonial past, regional proximity, and political alliances (Alesina and Dollar 2000). The grouping of countries by the nine economic regions are summarized in Table C1, Appendix C. In practice, although there are a small number of economies that belong in more than two economic communities (for instance, Angola is in both ECCAS and SADC), given the needs for the country inclusion to be mutually exclusive, we exercise discretion and include these economies in only one region. In addition, the lack of sufficiently long data series prevents us from analyzing Central Asia or Pacific Islands, even as a collective region, despite these fitting the context of our analysis too. The aggregate capital stock, K_t^C , refers to the sum of the private, public, and PPP capital stocks, and is obtained from the IMF *Investment and Capital Stock* dataset, whereas household consumption, C_t , is from the World Bank World Development Indicators. For consistency, we convert all the variables to be denominated in constant 2011 million international dollars. In sum, our dataset for estimation consists of **57** years of observations (losing one to first differencing) across nine economic regions, hence a panel of $57 \times 9 = 513$.

In line with the observation from Table 1 earlier (which appears to suggest difference stationarity), all the data are first-differenced to ensure stationarity. Following the guidance of Canova et al. (2012), we demeaned and standardize all the variables to avoid abnormalities of the variables with higher volatilities. Indeed, having also experimented with the detrended

data (not present to save space), it turns out that it does not matter we opt for first-differencing or detrending using Hodrick-Prescott filter as the results are qualitatively similar to those discussed in the next Section.

6 Results and Analysis

6.1 Benchmark - Sub-Saharan Africa-focused

For the benchmark empirical model (4×6), the estimated variance decomposition of selected variables are illustrated in Figure 5, with the detailed breakdown of the variance decomposition for all 24 variables (in response to one standard deviation shock on each of the variables) compiled in Table C2, Appendix C. For every variable, we examine up to a forecast horizon of 20 years. To begin, we first omit all the cross-regional feedback and spillover and focus only on intra-regional dynamics within the individual regions, notably the four African communities. Predictably, by design of the Cholesky ordering, for the two ‘relatively exogenous’ regions of developing Asia and LAC, at least 90 percent of the variations in loan aid flow, grant aid flow, and consumption are attributed to structural shocks within the region, even for up to a forward horizon of 20 years. Similar patterns are observed for the capital stock in the two regions, though there are larger feedback from other regions as we extend into a longer time horizon. For the four African regions of interest, a clear commonality can be observed: less than one percent of the variations in the loan aid and grant aid flows to all four regions are accounted for by shocks to consumption and capital stock, and this is stable regardless of the time horizon considered. This is true even for the case of traditional aid flows in ECCAS and ECOWAS, where only about 52 percent of the variations are explained by their own historical movements. This appears to be consistent with the finding of cross-country panel-based studies such as Burnside and Dollar (2000), which suggest that aid effectiveness is likely to be minimally influenced by good domestic policies and institutions in the recipient economies. Another commonality can be observed

in terms of the variation in consumption, where for all four African regions the traditional aid explains more of the variation than the loan aid. This is in direct opposite of the two ‘external’ regions (developing Asia and LAC), where loan aid is the relatively stronger contributor to consumption variation than traditional aid. In terms of the interaction between the two aid types, a notable difference across the four regions is observed: the co-movements of the two aid types appears to be strongest in the relatively poorer region of ECCAS, where shock to loan aid accounts for 13 percent of the variation in traditional grant aid, followed by SADC (5.7 percent), IGAD (3.25 percent), and then ECOWAS (1.07 percent). In terms of their combined contribution to the variation in aggregate capital stock (which is the closest measure for relative aid effectiveness in our empirical model), collectively for the economies in the IGAD region the two aid flows account for approximately 3.5 percent of the growth in capital stock, followed by ECCAS, ECOWAS, and SADC.

Indeed, if we were to examine now the contribution of shocks from outside the individual regions over the long run ($T = 40$), Table 4 summarizes the variance decomposition by the three categories of *Own region*, *Other Africa*, and *Outside Africa*. In the long run, notwithstanding the dynamics within the regions themselves, real consumption growth in the collective regions of SADC and ECCAS is influenced more by structural shocks from outside of Africa (with less than 10% spillover from other African regions), whereas for ECOWAS and IGAD the reverse is true. In terms of the growth rates of region-specific capital stock, except for the SADC region, we see that structural shocks from other African communities become a lot more significant, accounting for at least 28 percent of the variation in capital stock growth in the long run. Casting these observations in light of the objective of pursuing a single common market across the Sub-Saharan African continent, it appears that the supply-side progress of investment facilitation and the free movement of capital is far ahead of the demand-side initiative of final goods market integration.

To further infer on the effects of both types of aid on investment, we evaluate the impulse response functions (IRFs) in Figure 6, which presents the responses of aggregate capital

stocks (for all four African regions) to a series of one standard-deviation positive shocks simulated on the loan and grant aids in the different regions. Generally, except for capital investment in ECCAS, aggregate capital accumulation in the other three African regions would react negatively to a temporary increase in aid flows to the two non-African regions, i.e. a *substitution effect* between continents. For the ECCAS, capital accumulation appears to benefit not just from inflow of both aid types into its region (its instantaneous reaction to a rise in grant aid inflow is negative, but overall the effects are positive over the long run), but also positively to increases in aid flows in all the other regions, except ECOWAS. This appears to suggest that over the past 60 years investments in the ECCAS region tend to benefit when the global aid landscape is favorable to developing regions. Nevertheless, economies in the ECOWAS appear to be direct competitors for aid flow against the Central African economies, and a positive shock to both loan and grant aid flows into ECOWAS corresponds to a sharp drop in capital investment in the ECCAS region.

For the collective region of ECOWAS capital investment appears to be anti-cyclical to an increase in both aid types into the region. Indeed, the IRFs for capital stock in ECOWAS generally show negative reaction to positive shocks on all aid flows, except for when in response to aid flows into the IGAD region, as well as traditional aid into the ECCAS region. We believe the generally negative effects associated with increases in aid inflows into ECOWAS either reflect a ‘natural resource curse’-like effects (as in Djankov et al. 2008) or that they merely reflect the cyclical pattern of humanitarian aid: in period when the relative level of conflicts in one of the underdeveloped region is higher, as reflected in a positive shock to traditional aid, aggregate capital accumulation in said (the other) region naturally reduces (increases). Indeed, a closer inspection of the IRFs for capital stock in the IGAD and SADC regions in response to positive shocks to either aid type in ECOWAS suggests negative aid productivity in the collective ECOWAS region. This is similar in the SADC region. Despite housing the relatively rich economies of South Africa and Botswana, positive shocks to both aid types in SADC adversely affect capital accumulation in the region, although unlike

ECOWAS the region does benefit from spillover effect associated with positive shocks to the ECCAS and IGAD regions.

Similar to the ECCAS, the data of IGAD region appears to fit our proposed theoretical framework of a dual-objective donor-driven growth well too. Despite the Horn of Africa proximity, the IGAD group includes the fast-growing East African Community (EAC) duo of Kenya and Uganda, both of which are at the forefront of the continent in adopting the public-private partnership (PPP) framework to investment. On top of the largely positive spillover effect (to capital investment in other Sub-Saharan African regions) that has documented, aggregate capital accumulation in IGAD also reacts positively to an increase in both aid types flowing into the region, reflecting that the overall scale effect of an aid expansion outweighs the trade-off between loan and traditional grant aid. In combination, these therefore signal high aid productivity. In summary, casting our results against the general consensus from ‘aid and investment in Africa’ literature dated back to Dollar and Easterley (1999), Lancaster (1999)—the needs of a combination of good policies and aid to create a productive environment for private investment—it appears that the EAC, followed by the ECCAS, is aid productivity is at the front of the curve. In contrast, ECOWAS and SADC do not appear to display evidence of aid-driven growth, with positive shocks to both aid types flowing into the two regions signaling bad times across the continent.

6.2 Extension 1 - Latin America-focused

For the first of the two additional sets of estimation, we focuses more on the interaction of the three LAC regions of MERCOSUR, CAN, and CSME-SICA, as well as the linkages to the aggregate regions of Developing Asia and Africa. The variance decomposition of selected variables in the estimated 4×5 SVAR system are illustrated in Figure 7. Table 5 summarizes the variance decomposition by the three categories of *Own region*, *Other LAC*, and *Outside LAC* in the long run ($T = 20$), with the detailed breakdown of the variance decomposition for all 20 variables documented in Table C3, Appendix C. At first glance

of Table 5, overall the LAC region is less regionally integrated compared to Sub-Saharan Africa, as average spillover from other LAC regions to a single region of concern is merely 9.4 percent in the long run, which is much lower than the 18.2 percent observed earlier for Africa. Nevertheless, when we investigate the variance decomposition of individual variables further, some of the patterns observed for the African regions remain valid. For instance, in terms of commonalities, very little of the variations in the loan aid and grant aid flows to all three regions are explainable by shocks to domestic consumption and capital stock. In terms of consumption variation, notwithstanding that the majority of variation is explained by its past shocks, there is a difference between the two South American regions and CSME-SICA. Specifically, for the latter loan aid is a stronger predictor of consumption growth than traditional grant aid, accounting for the 7.8 percent of the variation, whereas for MERCOSUR and CAN the combination of both aid flows account for less than 4 percent of the variation in their respective consumption. In terms of the the dynamics of aggregate capital stock, all three LAC regions have shown limited evidence of a linkage between aid flows and capital investment. Within each of the three regions, even when both aids are combined with private consumption, they do not explain more than 3 percent of the growth in capital stock. Indeed, compared to Sub-Saharan Africa, the LAC regions appear to have much less regional integration in terms of investment flows, as except for CAN over half of the variation in capital stock in the other two regions is explained by its past shock. Even in the case of CAN, the majority of the variation is explained by structural shocks from outside of the LAC.

Again, to further investigate the fit of the dual-objective donor-driven growth framework, we examine impulse response functions (IRFs) of aggregate capital stock in the three LAC regions in response to aid shocks both within and from outside the regions. These are illustrated in Figure 8. For MERCOSUR, which includes some of the largest economies in the LAC region, we largely observe its capital accumulation to be negatively influenced by positive shocks to aid flows into other regions, consistent with the *substitution effect*

(between regions and continents) documented earlier. In terms of the response to aid flows into MERCOSUR economies, the IRFs are consistent to the mechanism of the theoretical model. Specifically, a temporarily rise in loan aid positively affects capital accumulation whereas the response to grant aid shock is negative (unlike observed for IGAD and ECCAS earlier, the total scale effect of aid expansion does not dominate the ‘traditional versus commercial’ aid trade-off). In contrast, for CAN the opposite is observed where capital stock responds positively to grant aid but negatively into the CAN region. Interestingly, investment dynamics in CAN responds positively to a shock on loan aid into the MERCOSUR region, which perhaps reflects some positive spillover effect from the larger economies of Argentina and Brazil to the smaller Andean economies. Lastly, except for the positive shock to grant aid in CAN, the IRFs for capital stock in the Central American and Caribbean regions largely displays the substitution effect, where positive shocks to aid flows in other regions largely result in negative effect to capital accumulation in the collective CSME-SICA region.

In sum, except for limited evidence in MERCOSUR, the evidence of a dual-objective donor-driven growth cycle is weaker in the LAC region, and this is perhaps reflected in the lower level of intra-regional integration compared to the Sub-Saharan African region examined earlier.

6.3 Extension 2 - Developing Asia-focused

For the final set of estimation, which dwells further into the interaction of the two developing Asian regions of ASEAN and SAARC (a 4×4 SVAR), the estimated variance decomposition of selected variables is illustrated in Figure 9, with the detailed breakdown of variance decomposition for all 16 variables documented in Table C4, Appendix C. In addition, Table 6 summarizes the variance decomposition by the three categories of *Own region*, the *Other Asian region*, and *Outside Asia* in the long run ($T = 20$), while the IRFs of aggregate capital stock in the two developing Asian regions in response to aid shocks are illustrated in Figure 10.

Although there is technically only two developing Asian regions that we are examining here, a previously documented pattern remains: the ‘domestic’ variables of consumption and capital stock account for less than 1.5 percent of the variation in both types of aid flows, which is very low even if by design this is attributable to the Cholesky ordering of variables. Similar to the LAC regions, inter-regional integration between ASEAN and SAARC also lags significantly behind Sub-Saharan Africa, though capital accumulation in both regions depends a lot more on structural shocks originated from the two continental regions outside of Asia. The collective ASEAN region, which includes Singapore and reasonably industrialized economies such as the Asian Tigers, has some interesting variance decomposition. Notably, only approximately 24 percent of its consumption variation is explained by its past shock—lowest among all the consumption variations we have examined, whereas shocks to loan aid account up to 33 percent. Our explanation for this, coupled with the IRFs, is that ASEAN as a whole has too high degree of openness and too many economies further along the income ladder to fit well into a framework of aid-driven growth cycle. On the contrary, despite its variance decomposition suggesting plenty of exogenous forces—beyond our model’s explanations—at play, the IRFs for the collective SAARC region appears to be consistent with the dual-objective donor-driven growth proposition. Specifically, despite an initial negative reaction to loan aid shock into the region, the long-run effects of positive shocks to both aid types are favorable to capital accumulation in the region—results that are consistent with the positive aid-GDP growth relationship documented by Asteriou (2009) for five South Asian economies. In addition, the IRFs also largely reflect the substitution effect too, in that capital accumulation in South Asia tend to reacts negatively to temporary aid expansion in other developing regions.

7 Concluding Remarks

Based on a novel theoretical framework with a dual-objective donor giving grant aid (driven by benevolent objective) and loan aid (profitability motive), as well as a PPP-like investment structure and multi-region feedback mechanism, we estimate a panel structural VAR model of traditional grant aid, loan aid, private consumption, and aggregate capital stock using data of nine economic communities/regions covering the 1961-2017 period. By allowing for cross-heterogeneities and interdependencies across regions, we identify empirical commonalities and differences of the aid landscape in different regions. First, in all three variants of our estimated system, we find structural shocks to domestic consumption and capital stock to explain very little of the variation in both aid types. Second, although the overall variance decomposition results suggest that the aid-investment link is weak, the preliminary result of a Granger causality from loan aid to physical capital investment is mostly supported by impulse response analysis, with its effectiveness differs across the different regions. Specifically, among the Sub-Saharan African blocs, loan aid appears to be most productive in the Kenya and Uganda-driven EAC region (and to a lesser extent, the ECCAS), where a temporary expansion in loan aid leads to capital expansion not only in its own region, but also positive spillover into other Sub-Saharan African regions in the long run. In contrast, aid productivity appears to be the weakest in ECOWAS and then SADC, where both the within- and spillover effects are negative. In light of our South-South aid-driven growth theoretical framework, we believe that this observation not only reflects the *substitution effect* of aids between regions and continents, but also the respective region's absorption capacity. Lastly, between the three developing South continents, we find deeper regional integration between the Sub-Saharan African regions than the collective LAC regions or the links between the two Southern Asian regions (Southeast Asia and South Asia), but it generally lags behind in terms of integration with the rest of the developing world outside of Africa.

In terms of potential future research avenue, first we acknowledge that, in practice there are generally a lot more nuances to consider in terms of aid modality, as issues such as

conditionality and the precise architecture can have significant influence on aid productivity. Indeed, a lot of aid disbursed had been wasted due to instability and conflict over the years, which are factors beyond the difference of donor ideology or motives. These are issues that we do not control for but worth exploring in the future. Second, there is natural limitation to macro-regional based studies like ours, in that, the policy implications are largely limited to ‘big picture’. Given the inherent heterogeneity of individual economies, a tedious but potentially fruitful exercise of repeating the same estimation of our SVAR system for individual countries may be worth considering. Third, as documented in studies such as Leigland (2018), a growing number of project aid arrangements in developing economies would assume a ‘blended’ form, through multi-donor project preparation facilities, such as the World Bank Group’s *Global Infrastructure Facility* or the *Africa50* of the African Development Bank. Aid driven by commercial objectives therefore goes beyond ODA gross loan, and would have a political economy angle that needs to be modelled too. These are therefore areas worth researching into in the future.

References

- Aguiar, M., & Gopinath, G. (2007). Emerging market business cycles: The cycle is the trend. *Journal of Political Economy*, 115, 69-102.
- Alesina, A., & Dollar, D. (2000). Who gives foreign aid to whom and why? *Journal of Economic Growth*, 5, 33-63.
- Arndt, C., Jones, S., & Tarp, F. (2015). Assessing foreign aid's long-run contribution to growth and development. *World Development*, 69, 6-18.
- Asteriou, D. (2009). Foreign aid and economic growth: New evidence from a panel data approach for five South Asian countries. *Journal of Policy Modeling*, 31, 155-161.
- Berthélemy, J.-C. (2006). Bilateral donors' interest vs. Recipients' development motives in aid allocation: Do all donors behave the same? *Review of Development Economics*, 10, 179-194.
- Blanchard, O., & Quah, D. (1989). The dynamic effects of aggregate demand and supply disturbances', *American Economic Review*, 79, 654-673.
- Bulir, A., & Hamann, A. (2003). Aid volatility: An empirical assessment. *IMF Staff Papers*, 50, 64-89.
- Burnside, C., & Dollar, D. (2000). Aid, policies, and growth. *American Economic Review*, 90, 847-868.
- Cadot, O., Fernandes, A., Gourdon, J., Matteo, A., & de Melo, J. (2014). Evaluating aid for trade: A survey of recent studies. *World Economy*, 37, 516-529.
- Canova, F., Ciccarelli, M., & Ortega, E. (2012). Do institutional changes affect business cycles? Evidence from Europe. *Journal of Economic Dynamics & Control*, 36, 1520-1533.
- Canova, F., & Ciccarelli, M. (2013). Panel vector autoregressive models: a survey. Working Paper Series 1507, European Central Bank.
- Chenery, H., & Strout, A. (1966). Foreign assistance and economic development. *American Economic Review*, 56, 719-733.
- De Lombaerde, P., & Mavrotas, G. (2009). Aid for trade, aid effectiveness and regional absorption capacity. In: De Lombaerde, P., Puri, L. (eds.), *Aid for Trade: Global and Regional Perspectives*. *World Report on Regional Integration*, 39-61.
- Dieppe, A., Ginepro, R., Han, J., Korhonen, I., & Lodge, D. (2018). The transition of China to sustainable growth—implications for the global economy and the euro area. Occasional Paper Series 206, European Central Bank.
- Djankov, S., Montalvo, J., & Reynal-Querol, M. (2008). The curse of aid. *Journal of Economic Growth*, 13, 169-194.
- Dollar, D., & Easterley, W. (1999). The search for the key: aid, investment and policies in Africa. *Journal of African Economies*, 8, 546-577.
- Dreher, A., Fuchs, A., Parks, B., Strange, A., & Tierney, M. (2016). Apples and dragon fruits: The determinants of aid and other forms of state financing from China to Africa. Working Papers 0620, University of Heidelberg.
- Dreher, A., Fuchs, A., Hodler, R., Parks, B., Raschky, P., & Tierney, M. (2019). Is favoritism a threat to Chinese aid effectiveness? A subnational analysis of Chinese development projects. CEPR Discussion Papers 13840.
- Dumitrescu, E., & Hurlin, C. (2012). Testing for Granger non-causality in heterogeneous panels. *Economic Modelling*, 29, 1450-1460.

- Easterley, W. (2006). *The White Man's Burden: Why the West's efforts to aid the rest have done so much ill and so little good*. Oxford University Press.
- Easterley, W. (2007). Was development assistance a mistake? *American Economic Review*, 97, 328-332.
- García-Cicco, J., Pancrazi, R., & Uribe, M. (2010). Real business cycles in emerging countries? *American Economic Review*, 100, 2510-2531.
- Gomanee, K., Girma, S., & Morrissey, O. (2005). Aid and growth in Sub-Saharan Africa: accounting for transmission mechanisms. *Journal of International Development*, 17, 1055-1075.
- Heinrich, T. (2013). When is foreign aid selfish, When is it selfless? *Journal of Politics*, 75, 422-435.
- Imbeau, L. (1988). Aid and ideology. *European Journal of Political Research*, 16, 3-28.
- Jones, S. (2015). Aid supplies over time: Addressing heterogeneity, trends, and dynamics. *World Development*, 69, 31-43.
- Juselius, K., Møller, N., & Tarp, F. (2014). The long-run impact of foreign aid in 36 African countries: Insights from multivariate time series analysis. *Oxford Bulletin of Economics and Statistics*, 76, 153-184.
- Kharas, H. (2007). The new reality of aid. *Brookings Blum Roundtable 2007*. Wolfensohn Center for Development at the Brookings Institution, Washington, DC.
- Kilby, C., & Dreher, A. (2010). The impact of aid on growth revisited: Do donor motives matter? *Economics Letters*, 107, 338-340.
- Kragelund, P. (2008). The return of non-DAC donors to Africa: New prospects for African development? *Development Policy Review*, 26, 555-584.
- Kumi, E., Ibrahim, M., & Yeboah, T. (2017). Aid, aid volatility and sectoral growth in Sub-Saharan Africa: Does finance matter? *Journal of African Business*, 18, 435-456.
- Lancaster, C. (1999). *Aid to Africa: So Much To Do, So Little Done*. University of Chicago Press.
- Leigland, J. (2018). Public-private partnerships in developing countries: The emerging evidence-based critique. *World Bank Research Observer*, 33, 103-134.
- Lin, J., & Wang, J. (2017). Traditional aid is ineffective for structural transformation. In: Lin, J., Wang, J. (eds.) *Going Beyond Aid Development Cooperation for Structural Transformation*. Cambridge University Press, Cambridge.
- Mekasha, T., & Tarp, F. (2013). Aid and growth: What meta-analysis reveals. *Journal of Development Studies*, 49, 564-583.
- Mioiu, C., & Reddy, S. (2010). Development aid and economic growth: A positive long-run relation. *Quarterly Review of Economics and Finance*, 50, 27-39.
- Morrissey, O. (1993). The mixing of aid and trade policies. *World Economy*, 16, 69-84.
- Moyo, D. (2009). *Dead Aid: Why Aid Is Not Working and How There Is a Better Way for Africa*. Allen Lane, London.
- Naim, M. (2009). Rogue Aid. *Foreign Policy*. Available at: <https://foreignpolicy.com/2009/10/15/rogue-aid/>.
- Pallage, S., & Robe, M. (2001). Foreign aid and the business cycle. *Review of International Economics*, 9, 641-672.
- Paulo, S., & Reisen, H. (2010). Eastern donors and western soft law: Towards a DAC donor peer review of China and India? *Development Policy Review*, 28, 535-552.

- Pedersen, K. (2001). The Samaritan's dilemma and the effectiveness of development aid. *International Tax and Public Finance*, 8, 93–103.
- Rajan, R., & Subramanian, A. (2008). Aid and growth: What does the cross-country evidence really show? *Review of Economics and Statistics*, 90, 643–665.
- Riddell, R. (1987). *Foreign Aid Reconsidered*. John Hopkins University Press, Baltimore.
- Riddell, R. (2007). *Does Foreign Aid Really Work?* Oxford University Press, Oxford.
- Roodman, D. (2007). The anarchy of numbers: Aid, development, and cross-country empirics. *World Bank Economic Review*, 21, 255–277.
- Sims, C. (1980). Macroeconomics and reality. *Econometrica*, 48, 1–48.
- Uhlig, H. (2005). What are the effects of monetary policy on output? Results from an agnostic identification procedure. *Journal of Monetary Economics*, 52, 381–419.
- van Dam, P., & van Dis, W. (2014). Beyond the merchant and the clergyman: Assessing moral claims about development cooperation. *Third World Quarterly*, 35, 1636–1655.
- Vijil, M. (2014). Aid for trade effectiveness: Complementarities with economic integration. *World Economy*, 37, 555–566.

This Working Paper presents work-in-progress. Not peer-reviewed

Table 1: Volatility, Autocorrelation, Cross-correlation of the Level and Growth Rate of Traditional Aid, Loan Aid, Capital Stock, and Consumption

Volatility		$\sigma(a^0)$	$\sigma(a^1)$	$\sigma(K^C)$	$\sigma(C)$	$\sigma(\Delta a^0)$	$\sigma(\Delta a^1)$	$\sigma(\Delta K^C)$	$\sigma(\Delta C)$
CSME-SICA	CARICOM & Central American Integration System	0.85 (.08)	0.55 (.05)	0.62 (.06)	0.74 (.07)	0.28 (.03)	0.35 (.03)	0.01 (.00)	0.04 (.00)
CAN	Andean Community of Nations	0.63 (.06)	0.81 (.08)	0.61 (.06)	0.59 (.05)	0.19 (.02)	0.81 (.08)	0.01 (.00)	0.03 (.00)
MERCOSUR	Southern Common Market	0.41 (.04)	0.75 (.07)	0.56 (.05)	0.61 (.06)	0.28 (.03)	0.48 (.05)	0.01 (.00)	0.09 (.01)
ECCAS	Economic Community of Central African States	0.69 (.06)	1.05 (.10)	0.60 (.06)	0.79 (.07)	0.35 (.03)	0.60 (.06)	0.02 (.00)	0.12 (.01)
ECOWAS	Economic Community of West African States	1.04 (.10)	0.72 (.07)	0.44 (.04)	1.07 (.10)	0.27 (.03)	0.62 (.06)	0.02 (.00)	0.08 (.01)
SADC	Southern African Development Community	1.38 (.13)	0.92 (.09)	0.48 (.05)	0.61 (.06)	0.25 (.02)	0.52 (.05)	0.01 (.00)	0.03 (.00)
IGAD	Intergovernmental Authority on Development	1.27 (.12)	0.46 (.04)	0.78 (.07)	0.94 (.09)	0.34 (.03)	0.35 (.03)	0.03 (.00)	0.07 (.01)
ASEAN	Association of Southeast Asian Nations	0.89 (.08)	1.18 (.11)	1.01 (.09)	1.05 (.10)	0.27 (.03)	0.52 (.05)	0.02 (.00)	0.11 (.01)
SAARC	South Asian Association for Regional Cooperation	0.50 (.05)	0.42 (.04)	0.89 (.08)	0.78 (.07)	0.25 (.02)	0.31 (.03)	0.02 (.00)	0.04 (.00)
Mean		0.85	0.76	0.56	0.80	0.28	0.51	0.02	0.07
Autocorrelation		$\rho(a^0, a^0_{t-1})$	$\rho(a^1, a^1_{t-1})$	$\rho(K^C, K^C_{t-1})$	$\rho(C, C_{t-1})$	$\rho(\Delta a^0, \Delta a^0_t)$	$\rho(\Delta a^1, \Delta a^1_t)$	$\rho(\Delta K^C, \Delta K^C_t)$	$\rho(\Delta C, \Delta C_t)$
CSME-SICA	CARICOM & Central American Integration System	0.913	0.762	0.945	0.940	-0.145	-0.367	0.778	0.047
CAN	Andean Community of Nations	0.942	0.190	0.941	0.940	0.025	-0.03	0.859	0.033
MERCOSUR	Southern Common Market	0.762	0.735	0.947	0.905	-0.268	-0.387	0.890	0.111
ECCAS	Economic Community of Central African States	0.832	0.786	0.940	0.927	-0.324	-0.156	0.897	-0.153
ECOWAS	Economic Community of West African States	0.910	0.387	0.929	0.951	-0.052	-0.300	0.908	-0.173
SADC	Southern African Development Community	0.917	0.738	0.935	0.946	0.015	-0.259	0.917	0.099
IGAD	Intergovernmental Authority on Development	0.897	0.715	0.934	0.942	-0.123	-0.309	0.938	-0.001
ASEAN	Association of Southeast Asian Nations	0.915	0.838	0.956	0.937	-0.113	-0.234	0.907	-0.035
SAARC	South Asian Association for Regional Cooperation	0.875	0.439	0.945	0.946	-0.190	-0.101	0.959	-0.127
Mean		0.885	0.621	0.941	0.937	-0.142	-0.239	0.895	-0.022

Notes: * Standard deviations are estimated using the maximum likelihood method of Marquardt/Quadratic Hill; terms in bracket denote standard errors.
 ** The traditional (a^0) and loan aid (a^1) series refer to the aggregate total grants and total loans provided by DAC member economies to countries in the respective regions.
 Sources: OECD Aid Data, IMF Capital Stock Dataset, World Bank World Development Indicators, Authors' calculations

Table 2: Contemporaneous correlations between economic regions/communities

Level of Traditional Aid										First Difference of Traditional Aid									
	CSME-SICA	CAN	MERCOSUR	ECCAS	ECOWAS	SADC	IGAD	ASEAN	SAARC		CSME-SICA	CAN	MERCOSUR	ECCAS	ECOWAS	SADC	IGAD	ASEAN	SAARC
CSME-SICA	1.000	0.878	0.209	0.739	0.783	0.867	0.818	0.773	0.749	CSME-SICA	1.000	0.363	-0.121	-0.085	-0.084	0.071	0.104	-0.097	0.199
CAN	0.878	1.000	0.462	0.787	0.794	0.834	0.830	0.773	0.794	CAN	0.363	1.000	0.208	0.110	-0.018	0.120	0.238	-0.060	0.233
MERCOSUR	0.209	0.462	1.000	0.095	0.058	0.128	0.144	0.076	0.348	MERCOSUR	-0.121	0.208	1.000	0.128	0.013	0.514	0.210	-0.372	0.185
ECCAS	0.739	0.787	0.095	1.000	0.914	0.888	0.886	0.773	0.683	ECCAS	-0.085	0.110	0.128	1.000	0.219	0.268	0.168	0.235	0.336
ECOWAS	0.783	0.794	0.058	0.914	1.000	0.958	0.958	0.958	0.625	ECOWAS	-0.084	-0.018	0.013	0.219	1.000	0.385	0.025	0.275	0.012
SADC	0.867	0.834	0.128	0.888	0.958	1.000	0.941	0.941	0.700	SADC	0.071	0.120	0.514	0.268	0.385	1.000	0.378	0.367	0.193
IGAD	0.818	0.830	0.144	0.886	0.908	0.930	1.000	0.867	0.784	IGAD	0.104	0.238	0.210	0.168	0.025	0.378	1.000	0.019	-0.023
ASEAN	0.773	0.772	0.102	0.858	0.940	0.941	0.867	1.000	0.563	ASEAN	-0.097	-0.060	0.372	0.235	0.275	0.367	0.019	1.000	0.099
SAARC	0.749	0.794	0.348	0.683	0.625	0.700	0.784	0.563	1.000	SAARC	0.199	0.233	0.185	0.336	0.012	0.193	-0.023	0.099	1.000
Level of Loan Aid										First difference of Loan Aid									
	CSME-SICA	CAN	MERCOSUR	ECCAS	ECOWAS	SADC	IGAD	ASEAN	SAARC		CSME-SICA	CAN	MERCOSUR	ECCAS	ECOWAS	SADC	IGAD	ASEAN	SAARC
CSME-SICA	1.000	0.186	-0.115	0.642	0.601	-0.497	0.416	0.355	0.120	CSME-SICA	1.000	0.293	0.081	-0.072	0.138	0.223	-0.026	0.180	0.133
CAN	0.186	1.000	0.687	0.030	-0.041	-0.259	0.036	-0.395	0.168	CAN	0.293	1.000	0.172	-0.238	0.035	0.222	-0.089	0.002	-0.058
MERCOSUR	-0.115	0.687	1.000	-0.261	0.041	-0.617	-0.073	-0.663	0.026	MERCOSUR	0.081	0.172	1.000	0.010	0.175	-0.227	0.055	-0.205	-0.005
ECCAS	0.642	0.030	-0.261	1.000	0.828	0.603	0.369	0.506	0.279	ECCAS	-0.072	-0.238	0.010	1.000	-0.033	-0.078	-0.119	0.090	0.089
ECOWAS	0.601	0.141	-0.041	0.828	1.000	0.527	0.515	0.437	0.433	ECOWAS	0.138	0.035	0.175	-0.033	1.000	0.067	0.223	0.056	-0.007
SADC	0.497	-0.259	-0.617	0.603	0.527	1.000	0.455	0.730	0.327	SADC	0.223	0.222	-0.227	-0.078	0.067	1.000	-0.197	0.122	0.045
IGAD	0.416	0.036	-0.073	0.369	0.515	0.455	1.000	0.107	0.099	IGAD	-0.026	-0.089	0.055	-0.119	0.223	-0.197	1.000	-0.065	0.060
ASEAN	0.355	-0.395	-0.663	0.506	0.437	0.730	0.107	1.000	0.346	ASEAN	0.180	0.002	-0.205	0.090	0.056	0.122	-0.065	1.000	0.155
SAARC	0.120	0.168	-0.005	0.279	0.433	0.327	0.099	0.346	1.000	SAARC	0.133	-0.058	-0.005	0.089	-0.007	0.045	0.060	0.155	1.000

Table 3: Pairwise Dumitrescu-Hurlin Panel Granger non-causality test results:

	Z-bar statistics	P-value
$H_0: \Delta a_t^D$ does not Granger-cause Δa_t^L . $H_1: \Delta a_t^D$ does Granger-cause Δa_t^L for at least one panel.	1.024	0.306
$H_0: \Delta a_t^L$ does not Granger-cause Δa_t^D . $H_1: \Delta a_t^L$ does Granger-cause Δa_t^D for at least one panel.	-0.140	0.889
$H_0: \Delta a_t^D$ does not Granger-cause ΔK_t^C . $H_1: \Delta a_t^D$ does Granger-cause ΔK_t^C for at least one panel.	0.929	0.353
$H_0: \Delta K_t^C$ does not Granger-cause Δa_t^D . $H_1: \Delta K_t^C$ does Granger-cause Δa_t^D for at least one panel.	-0.665	0.506
$H_0: \Delta a_t^D$ does not Granger-cause ΔC_t . $H_1: \Delta a_t^D$ does Granger-cause ΔC_t for at least one panel.	1.879	0.060*
$H_0: \Delta C_t$ does not Granger-cause Δa_t^D . $H_1: \Delta C_t$ does Granger-cause Δa_t^D for at least one panel.	2.920	0.004***
$H_0: \Delta a_t^L$ does not Granger-cause ΔK_t^C . $H_1: \Delta a_t^L$ does Granger-cause ΔK_t^C for at least one panel.	2.059	0.040**
$H_0: \Delta K_t^C$ does not Granger-cause Δa_t^L . $H_1: \Delta K_t^C$ does Granger-cause Δa_t^L for at least one panel.	-0.139	0.889
$H_0: \Delta a_t^L$ does not Granger-cause ΔC_t . $H_1: \Delta a_t^L$ does Granger-cause ΔC_t for at least one panel.	3.620	0.000***
$H_0: \Delta C_t$ does not Granger-cause Δa_t^L . $H_1: \Delta C_t$ does Granger-cause Δa_t^L for at least one panel.	0.659	0.510
$H_0: \Delta K_t^C$ does not Granger-cause ΔC_t . $H_1: \Delta K_t^C$ does Granger-cause ΔC_t for at least one panel.	-0.279	0.781
$H_0: \Delta C_t$ does not Granger-cause ΔK_t^C . $H_1: \Delta C_t$ does Granger-cause ΔK_t^C for at least one panel.	5.348	0.000***

Note: * denotes significance at 10% level, ** at 5% level, * at 1% level. A lag order of 2 is specified for every test.

Table 4: Variance decomposition by Geographical Sources in the Long Run (T=20)

Benchmark Empirical Model (Developing Asia-LAC-SADC-IGAD-ECCAS-ECOWAS)

Variables	Outside Africa	Other Africa	Own region
loan aid_SADC	16.1%	0.0%	83.9%
grant aid_SADC	25.0%	0.0%	75.0%
consumption_SADC	12.7%	0.2%	87.1%
capital_SADC	38.0%	16.5%	45.5%
loan aid_IGAD	10.2%	7.9%	81.9%
grant aid_IGAD	14.3%	11.1%	74.6%
consumption_IGAD	5.7%	10.6%	83.7%
capital_IGAD	31.0%	27.7%	41.3%
loan aid_ECCAS	16.3%	19.4%	64.3%
grant aid_ECCAS	17.8%	16.6%	65.6%
consumption_ECCAS	35.9%	7.3%	56.8%
capital_ECCAS	28.1%	35.9%	36.0%
loan aid_ECOWAS	13.0%	36.5%	50.6%
grant aid_ECOWAS	6.3%	40.0%	53.7%
consumption_ECOWAS	10.1%	23.5%	66.5%
capital_ECOWAS	37.6%	38.2%	24.2%
Average	19.9%	18.2%	61.9%

Table 5: Variance decomposition by Geographical Sources in the Long Run (T=20)

Extension 1 - Latin America-focused (Developing Asia-Africa-MERCOSUR-CAN-CSME-SICA)

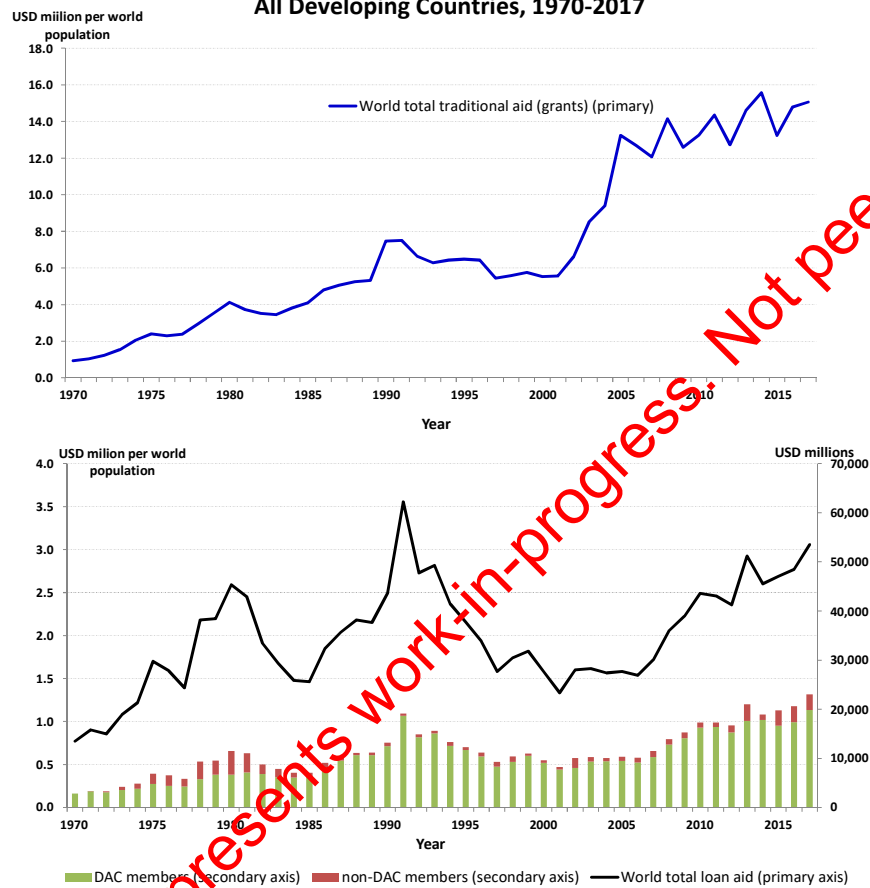
Variables	Outside LAC	Other LAC	Own region
loan aid_MERCOSUR	23.1%	0.1%	76.8%
grant aid_MERCOSUR	46.2%	0.2%	53.7%
consumption_MERCOSUR	14.4%	0.5%	85.1%
capital_MERCOSUR	30.6%	2.7%	66.7%
loan aid_CAN	12.4%		87.4%
grant aid_CAN	28.8%	1%	70.1%
consumption_CAN	12.8%	15.0%	72.1%
capital_CAN	55.7%	11.5%	32.8%
loan aid_CSME-SICA	8.8%	21.5%	69.7%
grant aid_CSME-SICA	28.0%	16.8%	55.2%
consumption_CSME-SICA	25.0%	23.3%	51.7%
capital_CSME-SICA	26.1%	14.8%	59.1%
Average	26.0%	9.4%	64.6%

Table 6: Variance decomposition by Geographical Sources in the Long Run (T=20)

Extension 2 - Developing Asia-focused (LAC-Africa-SAARC-ASEAN)

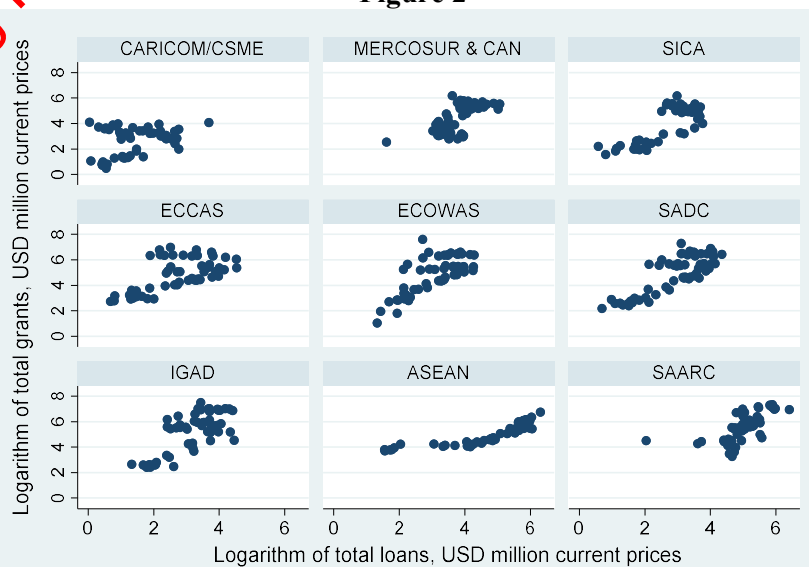
Variables	Outside Asia	Other Asian Region	Own region
loan aid_SAARC	3.8%	0.3%	95.9%
grant aid_SAARC	26.4%	0.2%	73.5%
consumption_SAARC	18.1%	0.5%	81.4%
capital_SAARC	49.0%	6.0%	44.9%
loan aid_ASEAN	6.1%	18.8%	75.1%
grant aid_ASEAN	26.9%	7.7%	65.4%
consumption_ASEAN	5.5%	37.2%	57.3%
capital_ASEAN	50.7%	4.4%	44.9%
Average	23.3%	9.4%	67.3%

Figure 1:
World Traditional Grant Aid (upper) & Gross Loan Aid (lower) to
All Developing Countries, 1970-2017



Sources: OECD Aid Data, Authors' calculation

Figure 2



Sources: OECD Aid Data, Authors' calculation

Figure 3

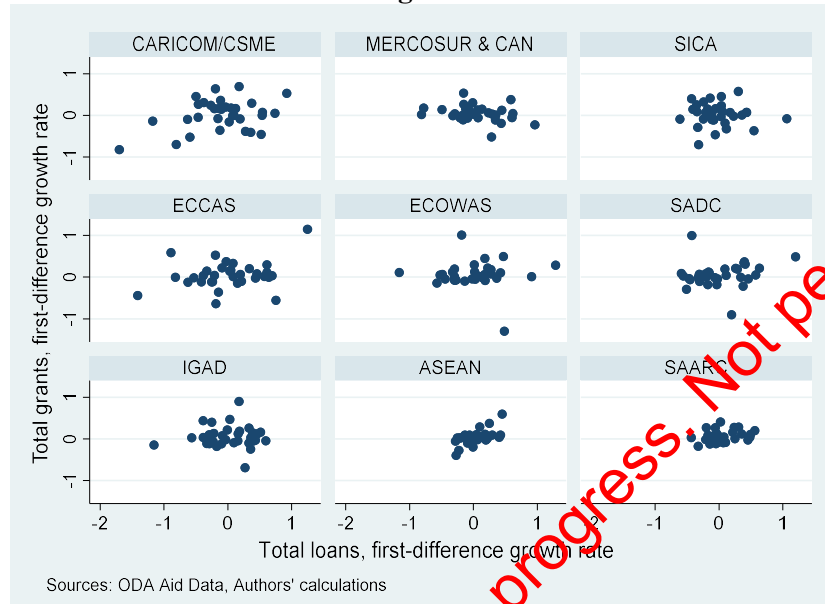


Figure 4

Loan Aid and PPP capital stock, 1960-2017 average

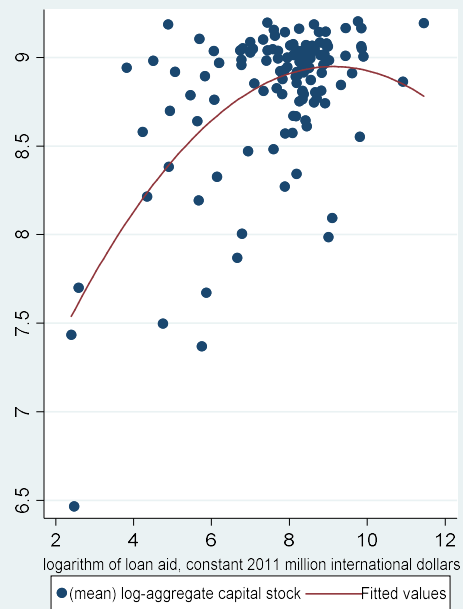
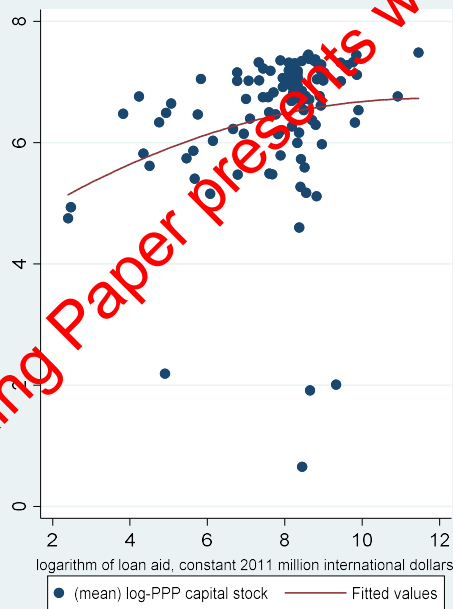


Figure 5: Variance Decoposition for for SADC, IGAD, ECCAS and ECOWAS

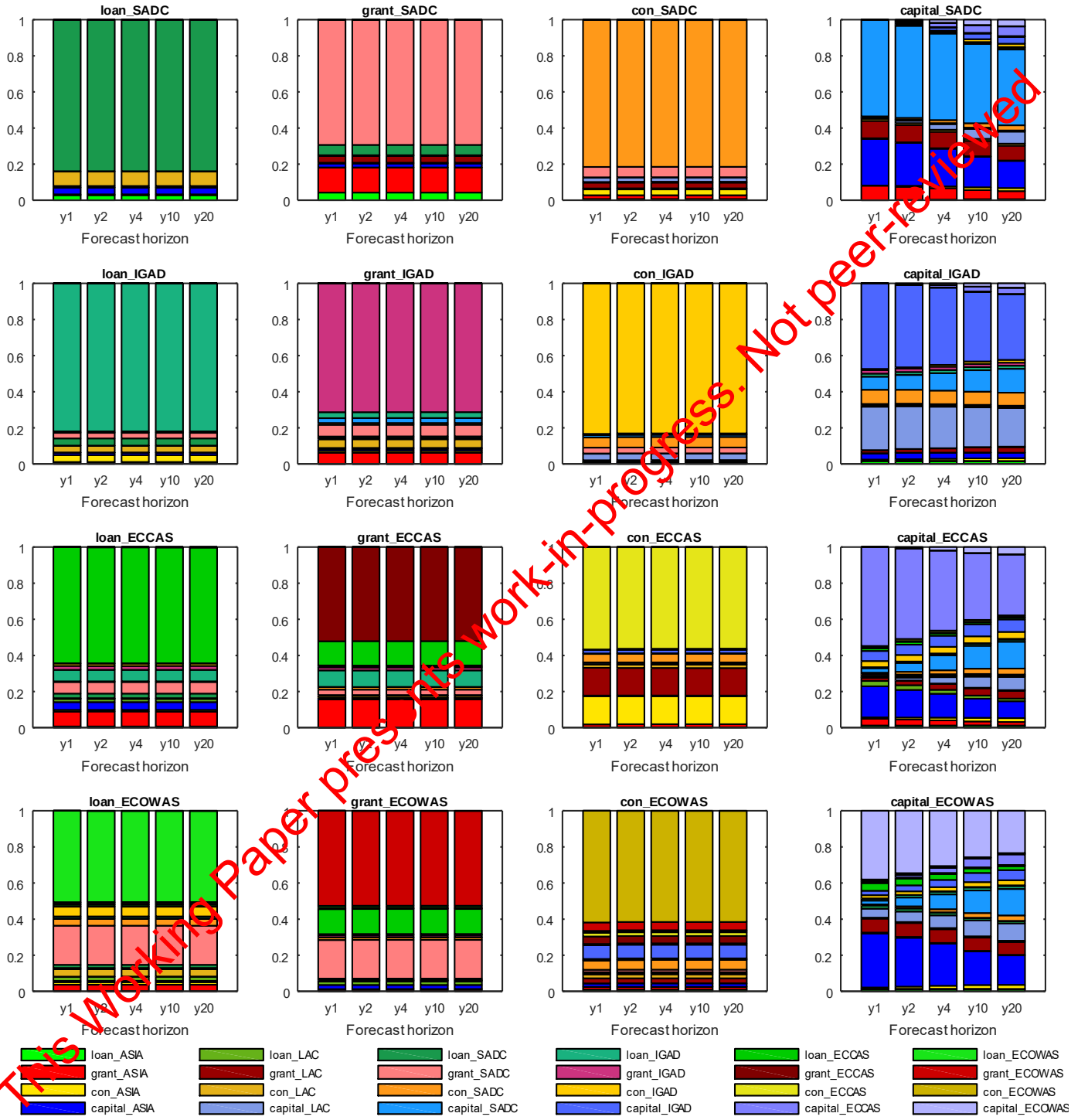


Figure 6: Impulse response functions for SADC, IGAD, ECCAS and ECOWAS

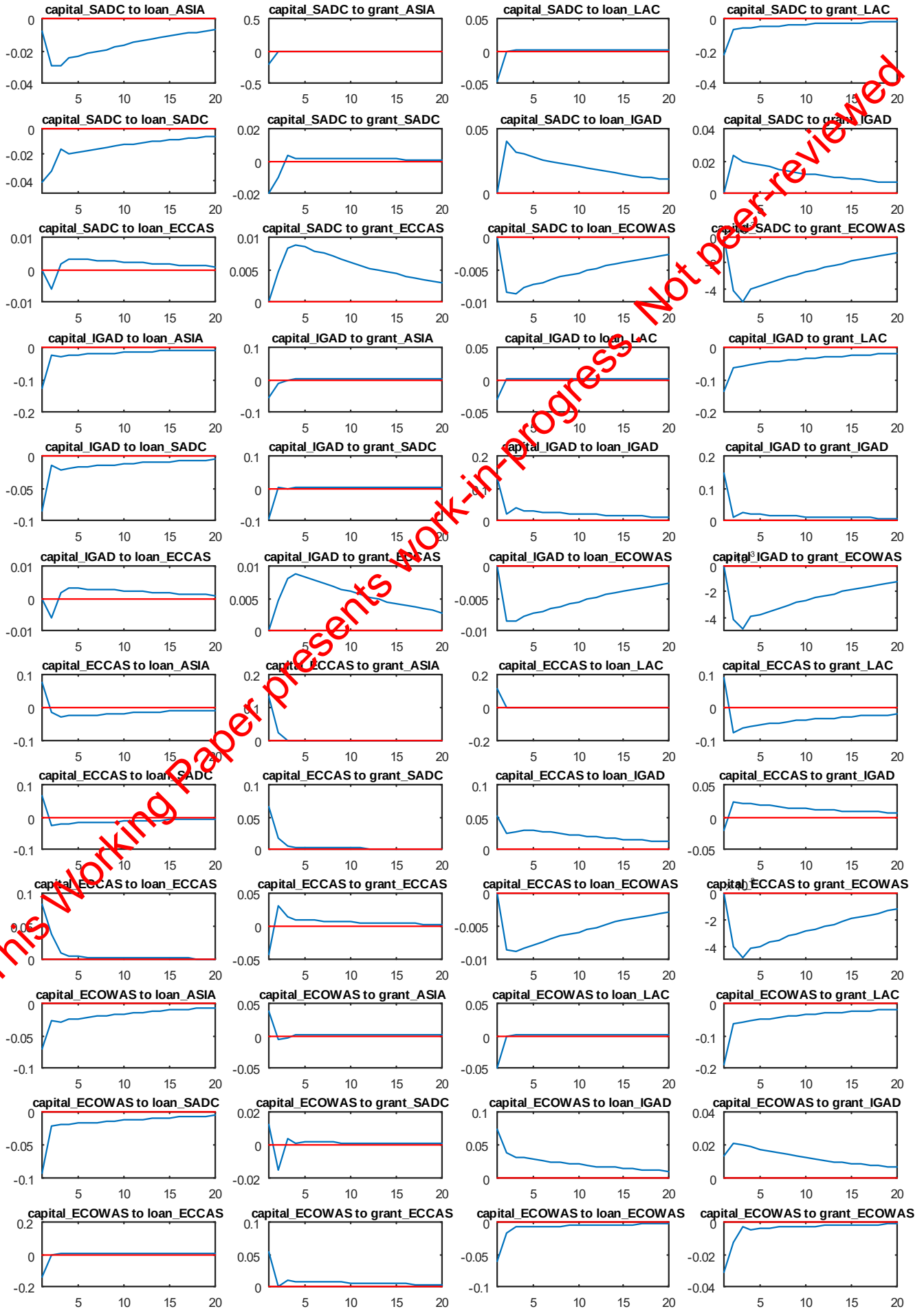


Figure 7: Variance Decoposition for MERCOSUR, CAN and CSME-SICA

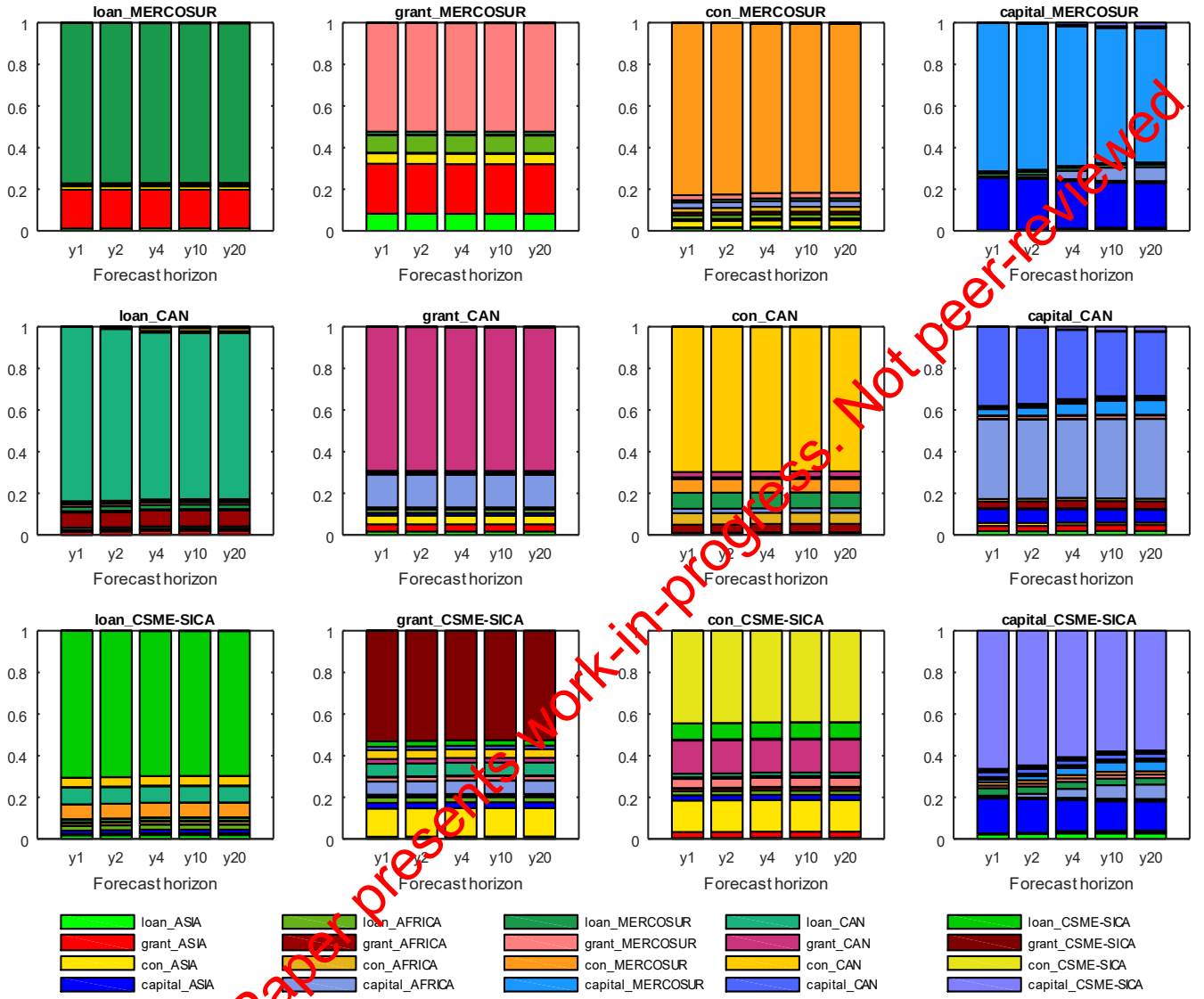


Figure 8: Impulse response functions for MERCOSUR, CAN and CSME-SICA

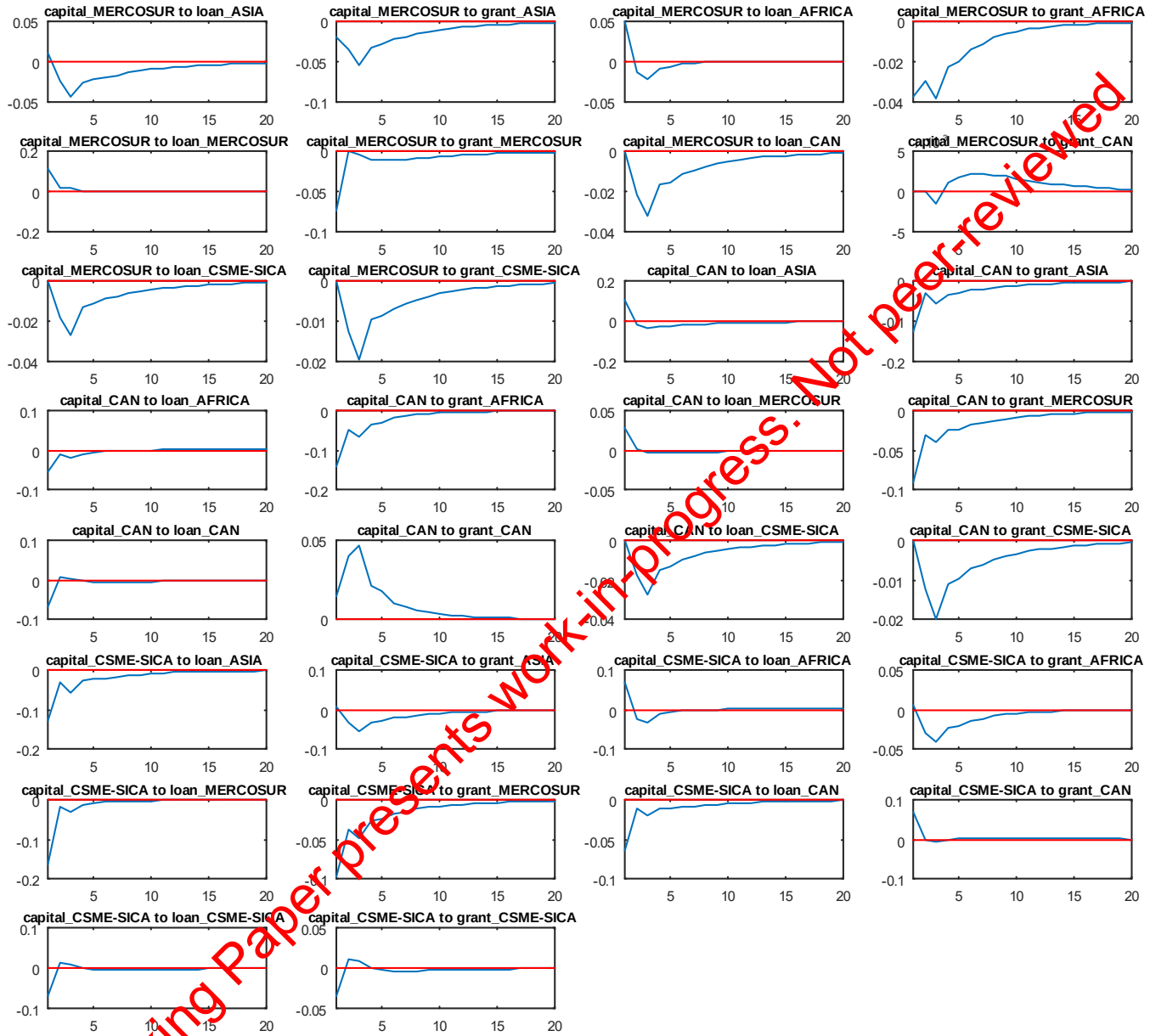


Figure 9: Variance Decomposition for SAARC and ASEAN

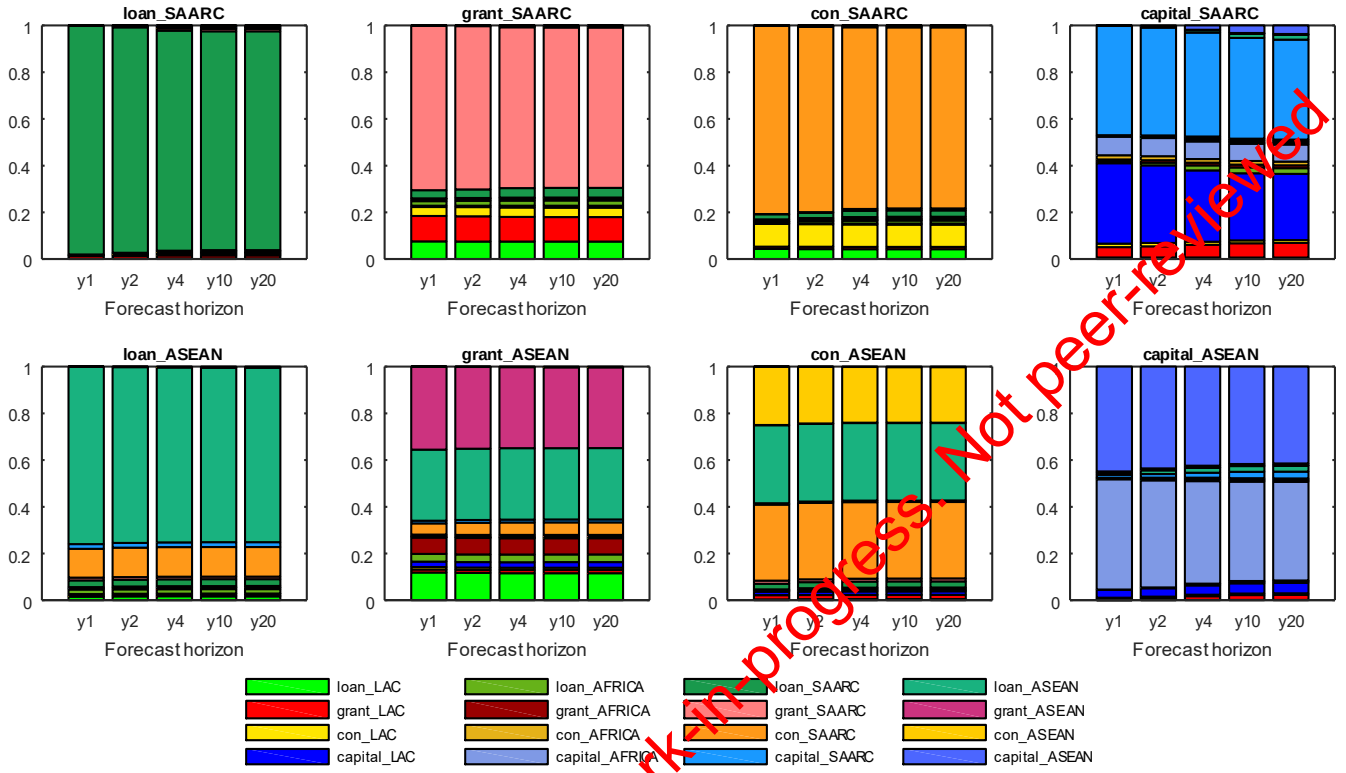
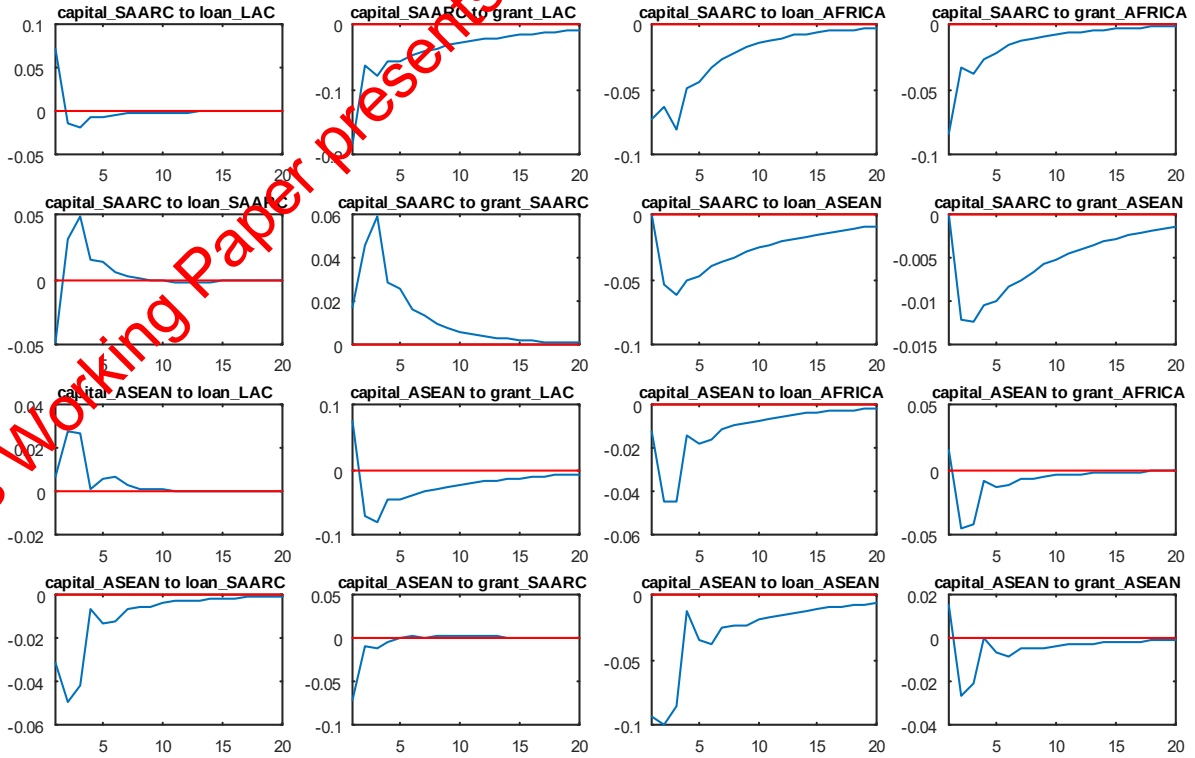


Figure 10: Impulse response functions for SAARC and ASEAN



DISCUSSION PAPERS IN ECONOMICS

- 2021/1 King Yoong Lim and Chunping Liu, *Dual-objective Donors and Investment across Economic Regions: Theory and Evidence*
- 2020/4 King Yoong Lim and Reagan Pickering, *Crime Heterogeneity and Welfare Spending: Theory and Empirical Evidence based on the Universal Credit System*
- 2020/3 Michael McCann, *Pension Funds and Socially-Responsible Investment in Corporate Debt Securities: An Empirical Investigation*
- 2020/2 King Yoong Lim and Shuonan Zhang, *Commodity Shocks and Optimal Fiscal Management of Resource Revenue in an Economy with State-owned Enterprises*
- 2020/1 King Yoong Lim and Diego Morris, *Learning Before and After the Global Crisis: Firm-level Innovation in Latin America.*
- 2019/5 Robert Ackrill and Rahmi Cetin, *The Impacts of Patent and R&D Expenditures on the High-Tech Exports of Newly Industrialised Countries: A Panel Cointegration Analysis.*
- 2019/4 Alan Collins, Jingwen Fan, and Aruneema Mahabir, *Using A LASSO Approach to Analyse the Determinants of Measured and 'Natural' Suicide Rates.*
- 2019/3 Leighton Vaughan-Williams, Chunping Liu, and Hannah Gerrard, *How Well Do Elo-based Ratings Predict Professional Tennis Matches?*
- 2019/2 Pengfei Jia and King Yoong Lim, *Police Spending and Economic Stabilization in a Monetary Economy with Crime and Differential Human Capital.*
- 2019/1 King Yoong Lim and Diego Morris, *Modeling the Drugs and Guns Trade in a Two-Country Model with Endogenous Growth.*
- 2018/3 Pengfei Jia and King Yoong Lim, *Tax Policy and Toxic Housing Bubbles in China.*
- 2018/2 Yousef Makhoul, *Trends in Income Inequality.*
- 2018/1 Dimitrios Bakas and Athanasios Triantafyllou, *The Impact of Uncertainty Shocks on the Volatility of Commodity Prices.*
- 2017/9 Eghosa Igudia, Rob Ackrill and Simeon Coleman, *Entrepreneurial Responses to Austerity: The Role of the Informal Sector.*
- 2017/8 Rahmi Cetin and Robert Ackrill, *Openness and Growth in Challenging Times: Analysing the Trade-Growth Nexus in Slovakia.*
- 2017/7 Dimitrios Bakas and Yousef Makhoul, *Can the Insider-Outsider Theory*

explain Unemployment Hysteresis in the OECD Countries?

- 2017/6 Dimitrios Bakas, Georgios Chortareas and Georgios Magkonis, *Volatility and Growth: A not so Straightforward Relationship*.
- 2017/5 Will Rossiter, *Prospects and Challenges for City Region Devolution in Nottingham and the East Midlands*.
- 2017/4 Chunping Liu and Zhirong Ou, *What determines China's housing price dynamics? New evidence from a DSGE-VAR*.
- 2017/3 Morakinyo O. Adetutu and Thomas G. Weyman-Jones, *Fuel subsidies versus market power: is there a countervailing second best welfare optimum?*
- 2017/2 Robert Mullings, *Do Institutions moderate Globalization's effect on Growth?*
- 2017/1 Anthony J Glass, Karligash Kenjegalieva, Victor Ajayi, Morakinyo Adetutu, Robin C. Sickles, *Relative Winners and Losers from Efficiency Spillovers in Africa with Policy Implications for Regional Integration*.
- 2016/4 Sara Ornati, *International Environmental Agreements and Domestic Politics*.
- 2016/3 Yanhui Zhu, Jingwen Fan and Jon Tucker, *The impact of UK monetary policy on gold price dynamics*.
- 2016/2 John Ebireri and Alberto Paloni, *Bank Development and a Lower Degree of Sophistication and Diversification of Developing Countries' Exports*.
- 2016/1 Robert Mullings and Aruneema Mahabir, *Growth by Destination: The Role of Trade in Africa's Recent Growth Episode*.
- 2015/1 Andrew Atherton, João R. Faria, Dongxu Wu and Zhongmin Wu, *Human Capital, Entrepreneurial Entry and Survival*.
- 2014/5 Leighton Vaughan Williams, *The US Supreme Court and the 'Affordable Care Act': An Exercise in Closed-Door Forecasting*.
- 2014/4 Juan Carlos Cuestas and Barry Harrison, *Unemployment Persistence in the EU15*.
- 2014/3 Francesco Marchionne and Sunny Parekh, *Growth, Debt, and Inequality*.
- 2014/2 Bo Jiang, Zhongmin Wu, Bruce Philp and Simeon Coleman, *Macro Stress Testing in the Banking System of China*.
- 2014/1 Eghosa Igudia, Rob Ackrill, Simeon Coleman and Carlyn Dobson, *Austerity Measures or Entrepreneurial Development? The case of the Nigerian Informal Economy*.
- 2013/5 Barry Harrison and Theodorus Wisnu Widjaja, *Did the Financial Crisis impact on the Capital Structure of Firms?*
- 2013/4 Geetha Ravishankar and Marie Stack, *The Gravity Model and Trade*

Efficiency: A Stochastic Frontier Analysis of Potential Trade.

- 2013/3 Chunping Liu and Patrick Minford, *How Important is the Credit Channel? An Empirical Study of the US Banking Crisis.*
- 2013/2 Chunping Liu and Patrick Minford, *Comparing Behavioural and Rational Expectations for the US Post-War Economy.*
- 2013/1 Dan Wheatley, *Is it good to share? Debating patterns in availability and use of job share.*
- 2012/3 Simeon Coleman and Vitor Leone, *Time-Series Characteristics Of UK Commercial Property Returns: Testing For Multiple Changes In Persistence.*
- 2012/2 Otavio Ribeiro de Medeiros and Vitor Leone, *Multiple Changes in Persistence vs. Explosive Behaviour: The Dotcom Bubble.*
- 2012/1 Rob Ackrill and Simeon Coleman, *Inflation Dynamics In Central And Eastern European Countries.*
- 2011/4 Simeon Coleman, *Inflation Dynamics and Poverty Rates: Regional and Sectoral Evidence for Ghana.*
- 2011/3 Dan Wheatley and Zhongmin Wu, *Work, Inequality, And The Dual Career Household.*
- 2011/2 Simeon Coleman and Kavita Sirichand, *Fractional Integration and the Volatility Of UK Interest Rates.*
- 2011/1 Simeon Coleman, *Investigating Business Cycle Synchronization In West Africa.*
- 2010/11 Marie Stack and Eric Pentecost, *A Gravity Model Approach To Estimating Prospective Trade Gains in The EU Accession And Associated Countries.*
- 2010/10 Vitor Leone And Bruce Philp, *Surplus-Value And Aggregate Concentration In The UK Economy, 1987-2009.*
- 2010/9 Robert Ackrill and Adrian Kay, *WTO Regulations and Bioenergy Sustainability Certification – Synergies and Possible Conflicts.*
- 2010/8 Paul Alagidede, Simeon Coleman and Juan Carlos Cuestas, *Persistence Of Inflationary Shocks: Implications For West African Monetary Union Membership.*
- 2010/6 Bruce Philp and Dan Wheatley, *The time scarcity and the dual career household: competing perspectives*
- 2010/5 Juan Carlos Cuestas, Sebastián Freille and Patricio O’Gorman, *The media and public agendas: testing for media effects in Argentina Turing the Kirchner administration*

- 2010/4 Vitor Leone, *From property companies to real estate investment trusts: the impact of economic and property factors in the UK commercial property returns*
- 2010/3 Juan Carlos Cuestas and Paulo José Regis, *Purchasing power parity in OECD countries: nonlinear unit root tests revisited*
- 2010/2 Juan Carlos Cuestas and Bruce Philp, *Exploitation and the class struggle*
- 2010/1 Barry Harrison and Winston Moore, *Nonlinearities in Stock Returns for Some Recent Entrants to the EU*
- 2009/7 Joao R. Faria, Le Wang and Zhongmin Wu, *Debts on debts*
- 2009/6 Juan Carlos Cuestas and Luis A. Gil-Alana, *Unemployment hysteresis, structural changes, non-linearities and fractional integration in Central and Eastern Europe*
- 2009/5 Juan Carlos Cuestas and Javier Ordóñez, *Unemployment and common smooth transition trends in Central and Eastern European Countries*
- 2009/4 Stephen Dobson and Carlyn Ramlogan, *Is there a trade-off between income inequality and corruption? Evidence from Latin America*
- 2009/3 Juan Carlos Cuestas and Luís Alberiko Gil-Alana, *Further evidence on the PPP analysis of the Australian dollar: non-linearities, structural changes and fractional integration*
- 2009/2 Estefanía Mourelle and Juan Carlos Cuestas, *Inflation persistence and asymmetries: Evidence for African countries*
- 2009/1 Juan Carlos Cuestas and Barry Harrison, *Further evidence on the real interest rate parity hypothesis in Central and Eastern European Countries: unit roots and nonlinearities*
- 2008/16 Simeon Coleman, *Inflation persistence in the Franc Zone: evidence from disaggregated prices*
- 2008/15 Juan Carlos Cuestas and Paulo Regis, *Nonlinearities and the order of integration of order prices*
- 2008/14 Peter Dawson and Stephen Dobson, *The influence of social pressure and nationality on individual decisions: evidence from the behaviour of referees*
- 2008/13 Juan Carlos Cuestas and Barry Harrison, *Testing for stationarity of inflation in Central and Eastern European Countries*
- 2008/12 Juan Carlos Cuestas and Dean Garratt, *Is real GDP per capita a stationary*

process? Smooth transitions, nonlinear trends and unit root testing

2008/11 Antonio Rodriguez Andres and Carlyn Ramlogan-Dobson, *Corruption, privatisation and the distribution of income in Latin America*

2008/10 Stephen Dobson and Carlyn Ramlogan, *Is there an openness Kuznets curve? Evidence from Latin America*

2008/9 Stephen Dobson, John Goddard and Frank Stähler, *Effort levels in contests: an empirical application of the Tullock model*

2008/8 Juan Carlos Cuestas and Estefania Mourelle, *Nonlinearities in real exchange rate determination: Do African exchange rates follow a random walk?*

2008/7 Stephen Dobson and John Goddard, *Strategic behaviour and risk taking in football*

2008/6 Joao Ricardo Faria, Juan Carlos Cuestas and Estefania Mourelle, *Entrepreneurship and unemployment: A nonlinear bidirectional causality?*

2008/5 Dan Wheatley, Irene Hardill and Bruce Philp, *"Managing" reductions in working hours: A study of work-time and leisure preferences in the UK industry*

2008/4 Adrian Kay and Robert Ackrill, *Institutional change in the international governance of agriculture: a revised account*

2008/3 Juan Carlos Cuestas and Paulo José Regis, *Testing for PPP in Australia: Evidence from unit root test against nonlinear trend stationarity alternatives*

2008/2 João Ricardo Faria, Juan Carlos Cuestas and Luis Gil-Alana, *Unemployment and entrepreneurship: A Cyclical Relation*

2008/1 Zhongmin Wu, Mark Baimbridge and Yu Zhu, *Multiple Job Holding in the United Kingdom: Evidence from the British Household Panel Survey*

DISCUSSION PAPERS IN POLITICAL ECONOMY

2006/3 Ioana Negru, *On Homogeneity and Pluralism within Economics Schools of Thought*

2006/2 David Harvie and Bruce Philp, *Learning and Assessment in a Reading Group Format or Reading Capital... For Marks*

2006/1 David Harvie, Bruce Philp and Gary Slater, *Regional Well-Being and 'Social Productivity' in Great Britain'*

2004/2 Massimo De Angelis and David Harvie, *Globalisation? No Question: Foreign Direct Investment and Labour Commanded*

2004/1 David Harvie, *Value-Production and Struggle in the Classroom, or, Educators Within, Against and Beyond Capital*

DISCUSSION PAPERS IN APPLIED ECONOMICS AND POLICY

2007/2 Juan Carlos Cuestas, *Purchasing Power Parity in Central and Eastern European Countries: An Analysis of Unit Roots and Non-linearities*

2007/1 Juan Carlos Cuestas and Javier Ordóñez, *Testing for Price Convergence among Mercosur Countries*

2006/2 Rahmi Cetin and Robert Ackrill, *Foreign Investment and the Export of Foreign and Local Firms: An Analysis of Turkish Manufacturing*

2006/1 Robert Ackrill and Adrian Kay, *The EU Financial Perspective 2007-2013 and the Forces that Shaped the Final Agreement*

2004/5 Michael A. Smith, David Paton and Leighton Vaughan-Williams, *Costs, Biases and Betting markets: New evidence*

2004/4 Chris Forde and Gary Slater, *Agency Working in Britain: Character, Consequences and Regulation*

2004/3 Barry Harrison and David Paton, *Do 'Fat Tails' Matter in GARCH Estimation? Stock market efficiency in Romania and the Czech Republic*

2004/2 Dean Garratt and Rebecca Taylor, *Issue-based Teaching in Economics*

2004/1 Michael McCann, *Motives for Acquisitions in the UK*

2003/6 Chris Forde and Gary Slater, *The Nature and Experience of Agency Working in Britain*

2003/5 Eugen Mihaita, *Generating Hypothetical Rates of Return for the Romanian Fully Funded Pension Funds*

2003/4 Eugen Mihaita, *The Romanian Pension Reform*

2003/3 Joshy Easaw and Dean Garratt, *Impact of the UK General Election on Total Government Expenditure Cycles: Theory and Evidence*

2003/2 Dean Garratt, *Rates of Return to Owner-Occupation in the UK Housing Market*

2003/1 Barry Harrison and David Paton, *The Evolution of Stock Market Efficiency in a Transition Economy: Evidence from Romania.*